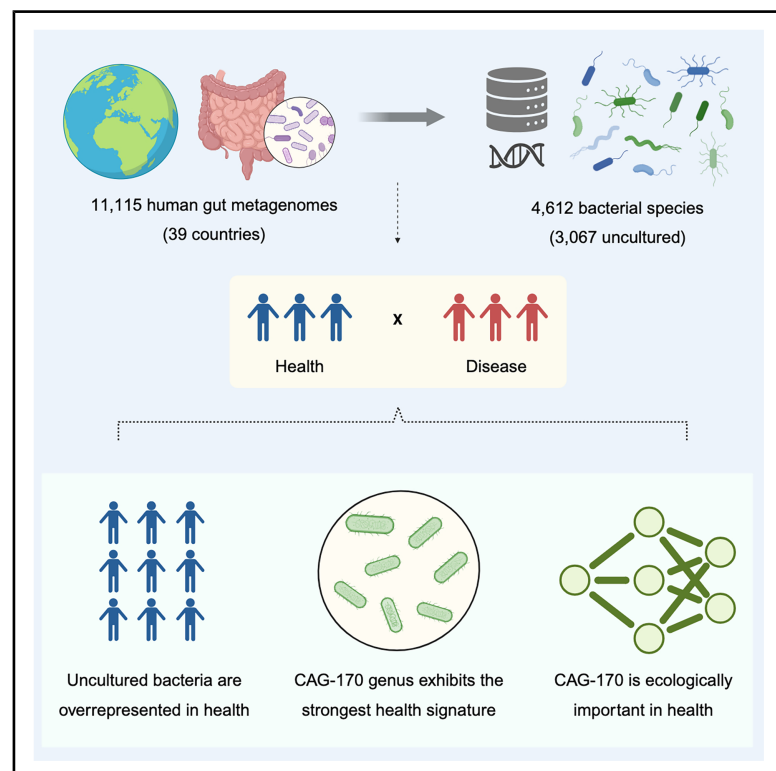


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Meta-analysis of the uncultured gut microbiome across 11,115 global metagenomes reveals a candidate signature of health

Graphical abstract



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In brief

Silva et al. perform a global analysis of over 11,000 gut microbiomes and reveal that uncultured bacteria are key markers of gut health. The uncultured genus CAG-170 is strongly linked to low gut dysbiosis and high vitamin B12 production and plays key ecological roles in the healthy gut.

Highlights

- Meta-analysis of 11,115 gut metagenomes links uncultured species to human health
- Healthy individuals show enrichment and greater diversity of uncultured bacteria
- The genus CAG-170 contains uncultured taxa with the strongest health signatures
- CAG-170 is ecologically important in healthy populations



Article

Meta-analysis of the uncultured gut microbiome across 11,115 global metagenomes reveals a candidate signature of health

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SUMMARY

The human gut microbiome is important for host health, yet over 60% of gut species remain uncultured and inaccessible to experimental manipulation. Here, we analyze 11,115 human gut metagenomes from 39 countries, 13 noncommunicable diseases, and healthy individuals to understand the clinical relevance of the uncultured microbiome worldwide. We identify 317 species linked to distinct clinical states, noting an overrepresentation of uncultured bacteria in healthy subjects. The genus CAG-170 emerged as the strongest health-associated lineage across multiple diseases and geographies, standing as the most central taxon based on ecological networks of healthy populations. We find that CAG-170 is temporally stable, with its abundance and subspecies diversity negatively correlated with gut imbalance over time. Functional predictions show CAG-170 species have greater vitamin B12 biosynthesis capacity and cross-feeding potential, providing important biological insights into this elusive genus. Our findings shed light on the underexplored role of uncultured gut species in health and disease.

INTRODUCTION

The human gut microbiome is a complex microbial ecosystem implicated in a variety of functions involved in host health, including nutrient metabolism, immune modulation, and colonization resistance against enteric pathogens.^{1,2} Bacterial cultivation of gut microbiome species has been an essential tool for studying the role of the human microbiome in health, as it enables the experimental manipulation and testing of bacterial isolates under controlled laboratory conditions.³ However, due to the complexity of the gut microbiome and the inherent challenges of culturing many of its constituent species, culture-independent methods, namely 16S rRNA gene sequencing and shotgun metagenomics, have been utilized to characterize microbiomes across different populations and health states.^{4,5}

A large body of research applying 16S rRNA gene sequencing to microbiome data has linked specific microbial signatures to health or diseases such as inflammatory bowel disease (IBD), colorectal cancer, and obesity, among others.⁶ For instance, members of the Oscillospiraceae and Ruminococcaceae families have been consistently implicated in health across diverse conditions.^{7,8} Although 16S rRNA genotyping has historically been the most widely used technique for microbiome analysis, it provides limited taxonomic and functional resolution due to its focus on a single marker gene. By contrast, metagenomics

offers a more comprehensive approach by sequencing the entire genetic material of a sample without targeted amplification.⁹ This leads to a higher taxonomic resolution, enabling the identification of species- and strain-level variations within microbial populations.¹⁰ Indeed, current metagenomic approaches have revealed more specific microbial signatures of health and disease across diverse populations and disease states, particularly in the context of IBD and colorectal cancer.^{11–14} One notable example was the discovery of a distinct subclade of the species *Fusobacterium nucleatum* found to be enriched in colorectal cancer patients.¹⁵ Furthermore, by characterizing the encoded genes and predicting their associated metabolic pathways, metagenomics can also uncover the functional potential of the microbiome community. For instance, a prior metagenomic study revealed that microbial pathways such as lipopolysaccharide biosynthesis and iron transport are associated with multiple disease conditions.¹⁶

However, developments in metagenome assembly and binning have revealed the existence of thousands of uncultured species in the human gut microbiome that remain poorly understood.^{17–19} Compared with cultured bacteria, uncultured species were found to have distinct metabolic profiles while also lacking specific biosynthetic pathways and antioxidant functions.^{17,19} These initial efforts culminated in the generation of the Unified Human Gastrointestinal Genome (UHGG) catalog,²⁰



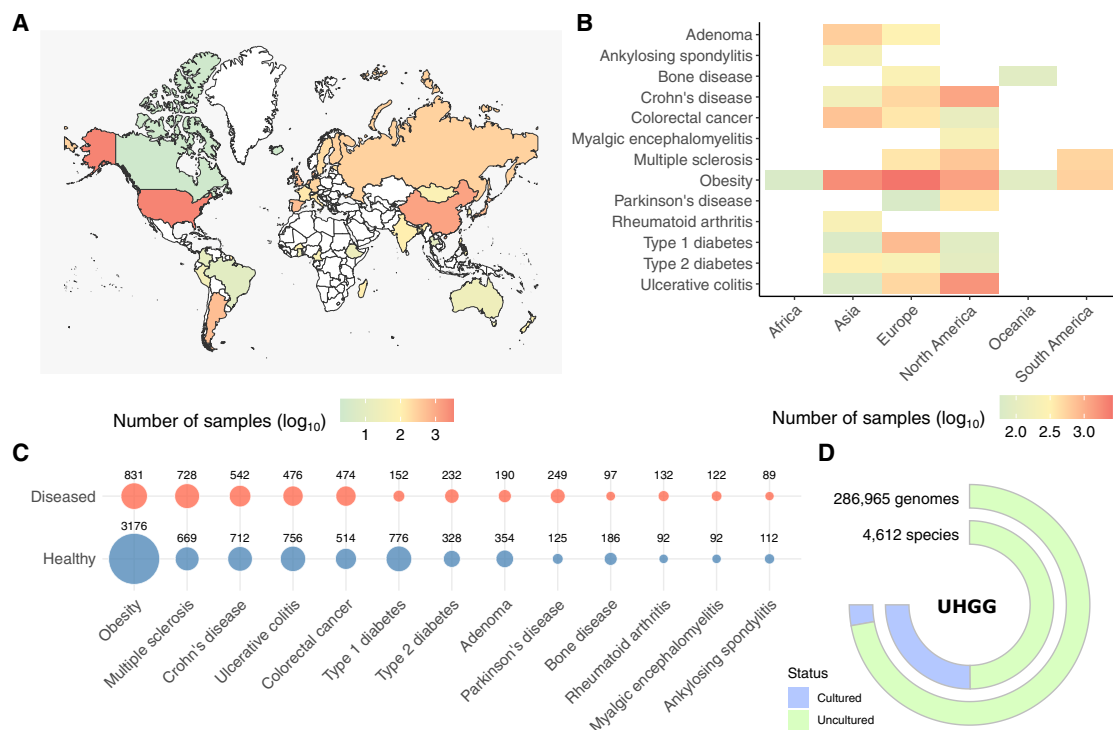


Figure 1. Mapping the uncultured gut microbiome across health and disease worldwide

(A) Geographic distribution of the 11,115 metagenomic samples analyzed in this study.

(B) Heatmap denoting the number of samples ($\log_{10}$ -transformed) processed per disease and per continent from case-control studies.

(C) Number of healthy and case (disease) metagenomic samples analyzed per condition.

(D) Distribution of the number of cultured (blue) and uncultured (green) genomes and species present in the Unified Human Gastrointestinal Genome (UHGG) catalog.

a sequence database compiling over 3,000 uncultured bacterial species representing a large diversity of taxa that cannot yet be experimentally characterized. This now offers unprecedented opportunities for accurate, high-resolution metagenomic analysis to perform a dedicated investigation of the role of the uncultured microbiome in health and identify priority candidates for developing targeted culturing approaches.

Here, we leveraged a dataset of over 11,000 human gut metagenomic samples across 13 noncommunicable diseases and healthy individuals to explore the clinical relevance of the uncultured gut microbiome. We discovered that the largely uncultured CAG-170 genus represents one of the strongest signatures of health and includes species with the highest centrality metrics in an ecological network analysis across healthy populations worldwide. Our study shows how genome-resolved metagenomics can shed light on the significance of understudied, as-yet uncultured members of the gut microbiome in host health.

RESULTS

Uncultured bacterial species are widespread across health and disease

To explore the clinical relevance of the uncultured microbiome in human health worldwide, we gathered 11,115 human gut metagenomic samples spanning 62 studies and 39 countries (Figure 1). In terms of geographical breadth, samples were

primarily collected from Europe (36%, $n = 3,845$), North America (29%, $n = 3,230$), and Asia (25%, $n = 2,755$), with most samples coming from the United States of America (29%), China (11%), and Denmark (6%).

We first focused our analysis on 8,672 human gut metagenomes sampled specifically from case-control studies across 13 noncommunicable diseases (Figure 1; Table S1). These included 4,358 samples (50.3%) from healthy controls, 1,682 samples (19.4%) from patients with gastrointestinal disorders (adenoma, colorectal cancer, Crohn's disease, and ulcerative colitis), 1,215 (14%) from metabolic diseases (obesity, type 1 and type 2 diabetes), 1,099 (12.7%) from neurological disorders (myalgic encephalomyelitis, multiple sclerosis, and Parkinson's disease), and 318 samples (3.7%) from other diseases such as ankylosing spondylitis, bone disease, and rheumatoid arthritis. After quality filtering and excluding samples with <500,000 reads, metagenomic samples represented a median of 25.8 million reads (interquartile range [IQR] = 9–46 million reads; Figure S1A).

Metagenomic samples were individually mapped to the UHGG catalog²⁰—a comprehensive reference database of genomic sequences from currently known prokaryotic species within the human gut microbiome. The UHGG catalog includes 4,612 species, of which 66.5% ($n = 3,067$) belong to prokaryotic species currently without a representative isolate genome (representing as-yet “uncultured” species; Figure 1D). To accurately quantify

the prevalence and abundance of each species in each sample, we employed a genome-based mapping approach that integrates metrics of both observed and expected genome breadth, along with depth of coverage, using previously validated thresholds.²¹ We obtained a median read mapping rate of 85.6% (IQR = 83.5%–87.4%) across all samples, which reduced to 61% (IQR = 57.4%–64.2%) after applying further filters based on breadth and depth of coverage (Figure S1B; see STAR Methods for further details).

A median of 187 prokaryotic species was detected per sample (IQR = 92–313 species), with 30.7% (IQR = 22–42%) belonging to uncultured bacteria. A total of 1,209 uncultured species were detected in at least 1% of samples, with the most prevalent species belonging to the genera *Faecalibacterium*, CAG-103, and CAG-170 (Figure S2A), the latter two representing genera currently without a cultured type strain. The proportion of uncultured species was generally consistent across health states (Figure S2B). However, in the context of Crohn's disease, ulcerative colitis, and obesity, the proportion of uncultured species was significantly higher in healthy controls compared with disease samples (linear mixed-effects model, false discovery rate [FDR] < 0.05). Regarding overall abundance, uncultured species represented a median of 11.8% of mapped reads (Figure S2B), indicating that although uncultured bacteria comprise ~30% of the species in the human gut, their relative abundance is lower when compared with cultured genomes.

Beyond calculating the number of species per sample, we estimated alpha diversity metrics using the Shannon index to account for both species richness and evenness. Based on the abundance of uncultured species alone, healthy controls showed significantly higher alpha diversity (linear mixed-effects model, FDR < 0.05) when compared with subjects from four diseases (ankylosing spondylitis, Crohn's disease, obesity, and ulcerative colitis; Figure S3A). Although the alpha diversity of cultured and uncultured bacteria generally correlated (Pearson $R^2 = 0.45$, $p < 0.0001$; Figure S3B), we detected statistically significant differences between the diversity of ulcerative colitis patients and healthy controls only when using the uncultured abundance profiles (linear mixed-effects model, FDR = 0.0066). Importantly, alpha diversity results were consistent across different normalization and rescaling strategies (Table S2). Overall, we show that uncultured bacteria are widely distributed across diverse health states, are disproportionately prevalent among certain healthy populations, and their diversity can better discriminate case and control samples in the context of ulcerative colitis.

Assessing the clinical relevance of the uncultured microbiome

Given the potential of the gut microbiome for disease prediction and diagnostic applications,^{22–24} we compared how well the composition of the cultured and uncultured microbiome could differentiate between healthy and diseased individuals. We evaluated three supervised machine learning approaches—ridge regression, random forest, and gradient boosting—across the 13 noncommunicable diseases tested and found that the choice of method was significantly associated with model performance (Kruskal-Wallis test, $p = 0.009$; Figure S4A). When using the uncultured microbiome alone, ridge regression yielded the highest

area under the receiver operating characteristic curve (AUROC) in 8 out of 13 diseases and a median AUROC of 0.728 (Figure S4B). This was followed by gradient boosting, which had the highest AUROC in three diseases, and random forest, which outperformed in two diseases. Consequently, all subsequent analyses were conducted using ridge regression to facilitate further comparisons.

On a per-disease basis (pooling subjects across all studies of one disease), models based on the uncultured microbiome performed best for ankylosing spondylitis (median AUROC = 0.916), ulcerative colitis (median AUROC = 0.814), and Crohn's disease (median AUROC = 0.778; Figure S5A). The performance of models using uncultured versus cultured species was slightly lower, with a median AUROC difference ($\text{AUROC}_{\text{uncultured}} - \text{AUROC}_{\text{cultured}}$) of -0.01 , but no statistically significant differences in most diseases (7 out of 13 diseases; Figure S5B). Since the uncultured microbiome represents a median of 12% of the microbial abundance detected in each sample, these results show this portion of the gut microbiome is nearly as clinically informative as the 88% represented by cultured bacteria. Notably, when comparing models trained on both uncultured and cultured bacteria to those using only cultured species, the addition of uncultured taxa significantly improved classification accuracy for ankylosing spondylitis and obesity (Figure S5C, Wilcoxon rank-sum test, FDR < 0.05).

We further investigated the generalizability of the machine learning models trained on both cultured and uncultured species using a cross-validation approach across different studies of the same disease (Figure S5D) and between different diseases (Figure S5E). When analyzing different studies of the same disease, gut-associated disorders (Crohn's disease, ulcerative colitis, and colorectal cancer) showed the greatest generalizability (≥ 0.7 AUROC in the cross-study validation), with type 1 diabetes, bone disease, and Parkinson's disease exhibiting the lowest performance (< 0.5 AUROC). Furthermore, in a cross-disease classification analysis, machine learning models were generally found to be disease specific, with limited cross-disease transferability (Figure S5E). However, a few notable exceptions were observed. The highest classification performances were obtained when models trained on myalgic encephalomyelitis were applied to Crohn's disease samples, when colorectal cancer models were tested on Crohn's disease, and when ulcerative colitis models were used to classify ankylosing spondylitis samples or vice versa (Figure S5E). Prior research has shown that Crohn's disease is a risk factor for both myalgic encephalomyelitis²⁵ and colorectal cancer,²⁶ while ulcerative colitis and ankylosing spondylitis are inflammatory disorders that share genetic predispositions and overlapping pathophysiological mechanisms.²⁷ These findings suggest that gut-associated disorders such as Crohn's disease, ulcerative colitis, and colorectal cancer have shared microbial features across different conditions that may represent more general markers of health and disease. Interestingly, the inclusion of the uncultured microbiome provided a significant benefit (Wilcoxon rank-sum test, FDR < 0.05) in 80% (24/30) of cross-disease comparisons with an AUROC > 0.6 (Figure S5F), further highlighting the relevance of the uncultured microbiome in certain clinical contexts.

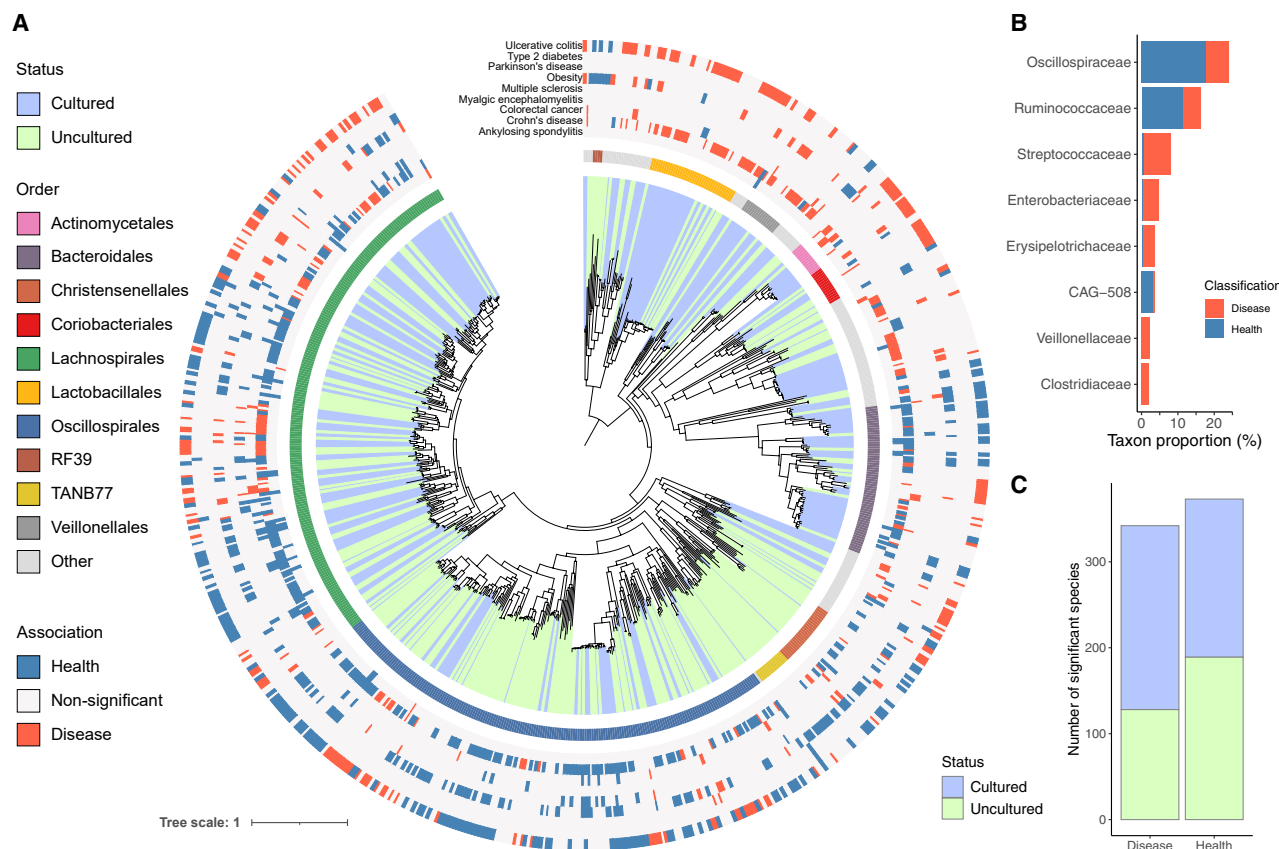


Figure 2. Phylogenetic structure of health- and disease-associated species

(A) Phylogenetic tree of the 713 bacterial species identified as associated with either health or disease. Outermost layers denote the association of each species across different conditions. The innermost layer indicates the taxonomic affiliation (order rank). Clades are colored based on the cultured status of the species. (B) Bacterial families found to be significantly associated with either health or disease (two-sided Fisher's exact test, FDR < 0.05). Taxon proportion represents the fraction of species that were linked to either health or disease within each family. (C) Number of cultured and uncultured species found as either health- or disease-associated. The proportion of uncultured species was found to be significantly higher (two-sided χ^2 test, $p = 0.00049$) among the health-associated taxa.

The genus CAG-170 is the strongest health-associated uncultured taxon

To identify microbial signatures of health and disease, we performed a differential abundance analysis using two complementary statistical methods based on generalized linear models^{28,29} (see STAR Methods). After accounting for the subject's age group, continent of origin, sample read depth, and batch (that is, study source), we identified 715 prokaryotic species associated with at least one disease (398 cultured, 317 uncultured; Figure 2A; Table S3). The number of significant species varied by disease, with most species found to be associated with Crohn's disease ($n = 414$), ulcerative colitis ($n = 396$), obesity ($n = 227$), and/or colorectal cancer ($n = 60$). Given the effect size and direction of association across the 13 diseases tested, we calculated a combined effect size to denote the overall strength of association of each species with either health (i.e., the control population) or disease. Out of 3,091 species detected at >1% prevalence in at least one case-control cohort, 342 species were classified as disease associated (overrepresented in disease; Figure S6A), whereas 373 species were health associated (overrepresented in the healthy controls; Figure S6B).

Of note, 74% of the disease-associated species were uniquely associated with one disease, whereas 56% of the health-associated species were consistent across multiple diseases, suggesting a more conserved signal of health. Only 25 species (3%) displayed conflicting associations—linked to health in some disease cohorts and to disease in others—indicating that most species exhibit consistent associations across different conditions.

At a taxonomic level, the families Oscillospiraceae, Ruminococcaceae, and CAG-508 were found to be significantly overrepresented among the health-associated species (Fisher's exact test, FDR < 0.0001; Figure 2B), whereas Streptococcaceae, Enterobacteriaceae, Erysipelotrichaceae, Veillonellaceae, and Clostridiaceae were overrepresented in disease. This signal was found to be consistent at higher taxonomic ranks, with the order Oscillospirales most strongly associated with health and the order Lactobacillales most overrepresented in disease.

We discovered that the proportion of uncultured species was significantly higher among the health-associated taxa compared with the disease-associated species (health: 184 cultured and 189 uncultured species; disease: 214 cultured and 128

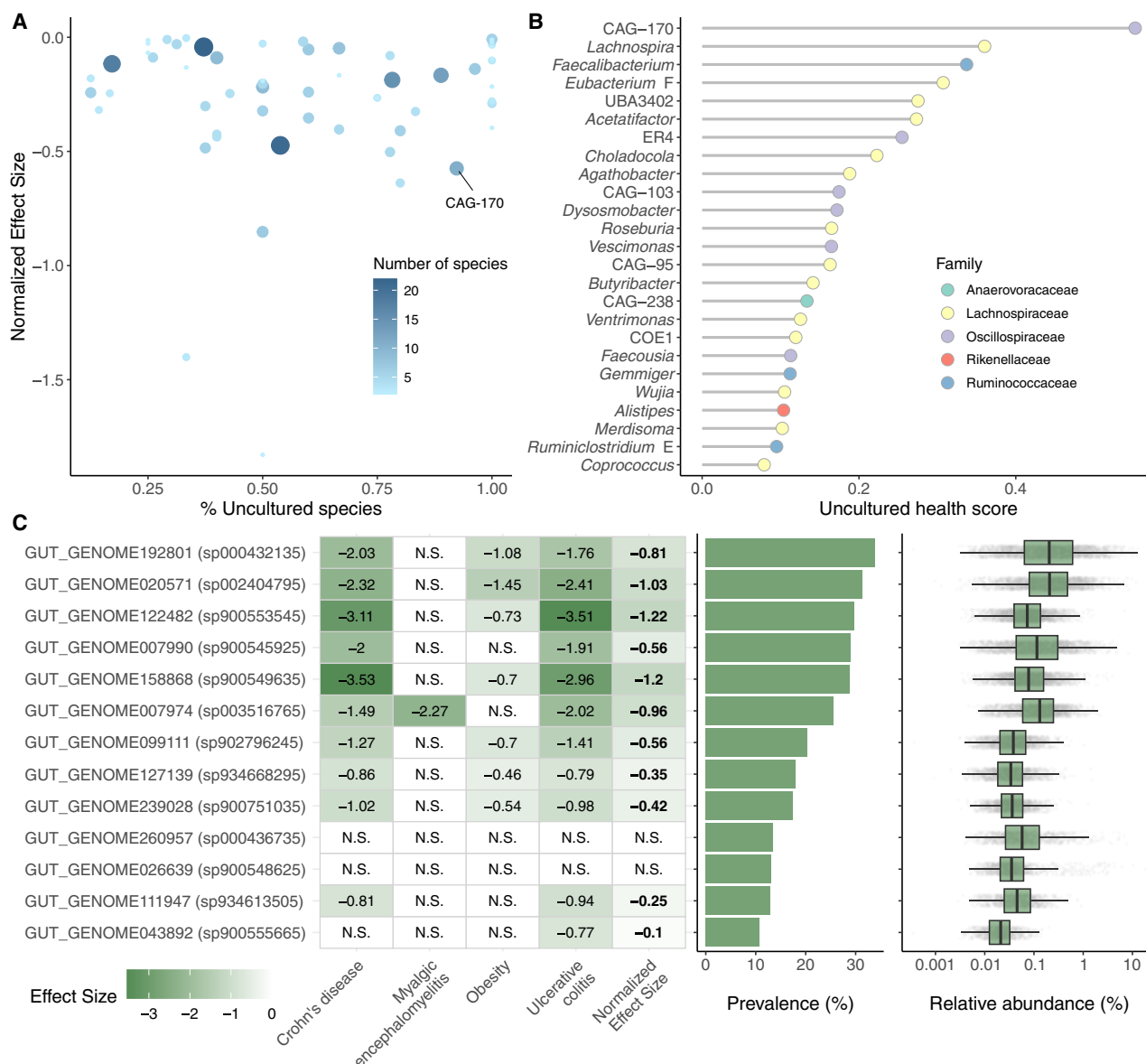


Figure 3. CAG-170 is the top-ranking taxon based on the number of uncultured species associated with health

(A) Proportion of uncultured species relative to the estimated effect size of each bacterial genus in the UHGG catalog. A negative effect size indicates an association with health. The size and color of each data point (genus) is scaled according to the number of health-associated species identified (only genera with an overall association with health are shown).

(B) Bacterial genera ranked by the absolute value of their uncultured health score, calculated as the product of the number of significant species ($\log_{10}$ -transformed), normalized effect size, and proportion of uncultured species.

(C) Effect sizes, prevalence, and relative abundance of the 13 species of CAG-170 present in the UHGG catalog. Values labeled as N.S. indicate the species was not significant in the specific disease cohort (FDR > 0.05). In the boxplots, the center line represents the median. Whiskers extend to the furthest point within 1.5 times the IQR from the box. The number of data points in the boxplots corresponds to the species prevalence among 8,672 samples.

uncultured species; χ^2 test, $p = 0.00049$), implicating the uncultured microbiome as a potential signature of health (Figure 2C). To investigate this further, we aimed to identify the specific uncultured taxa that most strongly underlined a healthy state. We therefore ranked all bacterial genera present in the UHGG catalog based on a combined measure that incorporated three key metrics: (1) the total number of health-associated species detected per genus, (2) a genus-level effect size normalized across

diseases and species, and (3) the proportion of uncultured species currently known within each genus (Figure 3A; Table S4). Using this combined measure, we determined that the unclassified genus CAG-170 was the top-ranking taxon (Figure 3B). This result was found to be consistent when aggregating effect sizes using an independent meta-analysis approach (Figure S7; see STAR Methods for further details). In fact, CAG-170 was the only genus that matched all the following criteria: (1) >10

significant health-associated species, (2) absolute effect size >0.5 , and (3) $>75\%$ uncultured species (Table S4). The taxon CAG-170 is a largely unknown bacterial genus from the Oscillospiraceae family, represented by 1,046 genomes and 13 species in the UHGG catalog. Since only one isolate genome from CAG-170 has been sequenced thus far,³⁰ while $>1,000$ metagenome-assembled genomes (MAGs) are available, this genus remains almost exclusively represented by uncultured genomes. Within our dataset, CAG-170 species were detected at a median prevalence of 20.3% (IQR = 13.4%–29.1%) and a median relative abundance of 0.07% (IQR = 0.03%–0.18%; Figure 3C). In the differential abundance analysis, 11/13 (85%) of CAG-170 species were found to be associated with health in the context of Crohn's disease, myalgic encephalomyelitis, ulcerative colitis, and obesity (Figure 3C). Collectively, these results show that CAG-170 has a consistent health-associated signal across multiple conditions and multiple species.

CAG-170 is the most interconnected genus within the global healthy microbiome

Having identified uncultured health-associated species through the analysis of case-control cohorts, we sought to define the broader characteristics of the healthy uncultured microbiome on a global scale. To achieve this, we aggregated all healthy control samples from the case-control studies analyzed in this work ($n = 3,614$) and incorporated an independent dataset comprising an additional 2,443 metagenomes from healthy individuals (Table S1). We first examined prevalence patterns on a per-continent basis (across Africa, Asia, Europe, North America, Oceania, and South America) to assess the presence of a dominant core microbiome either within or across regions. Prevalence was calculated as the proportion of samples in which each species was detected. Cultured species were found to be the most prevalent in Asia (median prevalence across species = 9.7%; IQR = 3.3%–31%), whereas uncultured species had their highest prevalence in Africa (median = 9.2%; IQR = 1.7%–23.1%). Across the full spectrum of gut species, we identified 29 cultured and 2 uncultured species within the top 1% most prevalent species ($>70\%$ prevalence) across the six continents analyzed (here defined as the “core” set of microbiome species; Figure 4A). When averaging species prevalences across continents, only one cultured species (*Blautia wexlerae*) reached a mean worldwide prevalence $>90\%$ among healthy individuals (Figure 4A), further emphasizing that most microbiome species are not universally conserved across diverse global populations. The two core uncultured species belonged to the *Faecalibacterium* genus and included an unclassified species detected in 77% of samples.

To complement the prevalence analysis, we further explored putative interspecies interactions of the healthy microbiome by modeling co-abundance patterns using a method based on SparCC,³¹ which specifically accounts for the compositional nature of microbiome data. Similar to the prevalence analysis, we constructed co-abundance networks separately for samples from each continent. From these networks, we computed three centrality measures for each species—closeness, betweenness, and degree—to capture different aspects of their influence on the overall network structure. From an ecological perspective, betweenness centrality quantifies how often a species acts as

a bridge or mediator between other taxa; closeness centrality reflects how close a species is to all others in the network, indicating its potential to rapidly influence or be influenced by the rest of the community; and degree centrality represents the number of direct connections a species has (i.e., how frequently it co-occurs with other taxa). Prior simulations have shown that keystone species—those with a disproportionate role in community stability—tend to exhibit high degree, high closeness centrality, and low betweenness centrality in co-occurrence networks.^{32,33} In fact, it was shown that models combining these three metrics could identify keystone taxa with 85% accuracy.^{32,33} Accordingly, we integrated these metrics into a composite centrality score averaged across continents and classified the species ranking in the top 1% centrality scores as network hubs. This revealed a total of 31 hub species, of which 27 were classified as uncultured (Figure 4B). Importantly, the CAG-170 genus was the most represented taxon among this set, with 12 species in the top 1% of the most central nodes. Even after filtering network edges with two different correlation thresholds, CAG-170 remained the most frequent genus identified among the top network hubs (Figure S8).

Lastly, combining all these results, we assessed the overlap among species classified as health-associated, disease-associated, core, and hubs to determine whether there were consistent patterns (Table S5). Although prevalence and centrality scores were found to generally correlate (Spearman's $\rho = 0.788$, $p < 0.0001$; Figure 4C), there was no overlap between species that were classified as core and network hubs (Figure 4D). In fact, for species with a mean global prevalence $>30\%$, there was no significant correlation between centrality scores and prevalence (Spearman's $\rho = 0.087$, $p = 0.195$). With regard to health state, most core and hub species were identified as health associated (Figure 4D), suggesting that such species are typically depleted in disease states. Uncultured bacteria were more frequent among species that were both hubs and health associated, with the genus CAG-170 again observed as the most represented within this group (Figure 4D).

We further examined the species exhibiting positive or negative associations with CAG-170 to gain further insights into its connection with the rest of the gut microbiome (Figure S9). After averaging correlation signals across continents and species, we found that positive interspecies associations with CAG-170 were stronger but less common. Specifically, 704 species were positively associated with CAG-170, compared with 2,280 negatively associated taxa. However, the top 10% of positively associated species showed a stronger correlation (higher effect size) than the top 10% of negatively associated species. CAG-170 had the highest positive correlation primarily with other uncultured species from the Oscillospirales order, while its strongest negative associations spanned a broader phylogenetic range, including members of the Lachnospirales, Bacteroidales, and Peptostreptococcales orders (Figure S9). Health-associated species were more common among the positive taxa, while disease-associated species were more frequent within the negative set (Figure S9C). However, 53% and 79% of the top positive and negative species, respectively, had no known health associations (Figure S9C). Together, these findings show that the CAG-170 genus is the most interconnected genus within

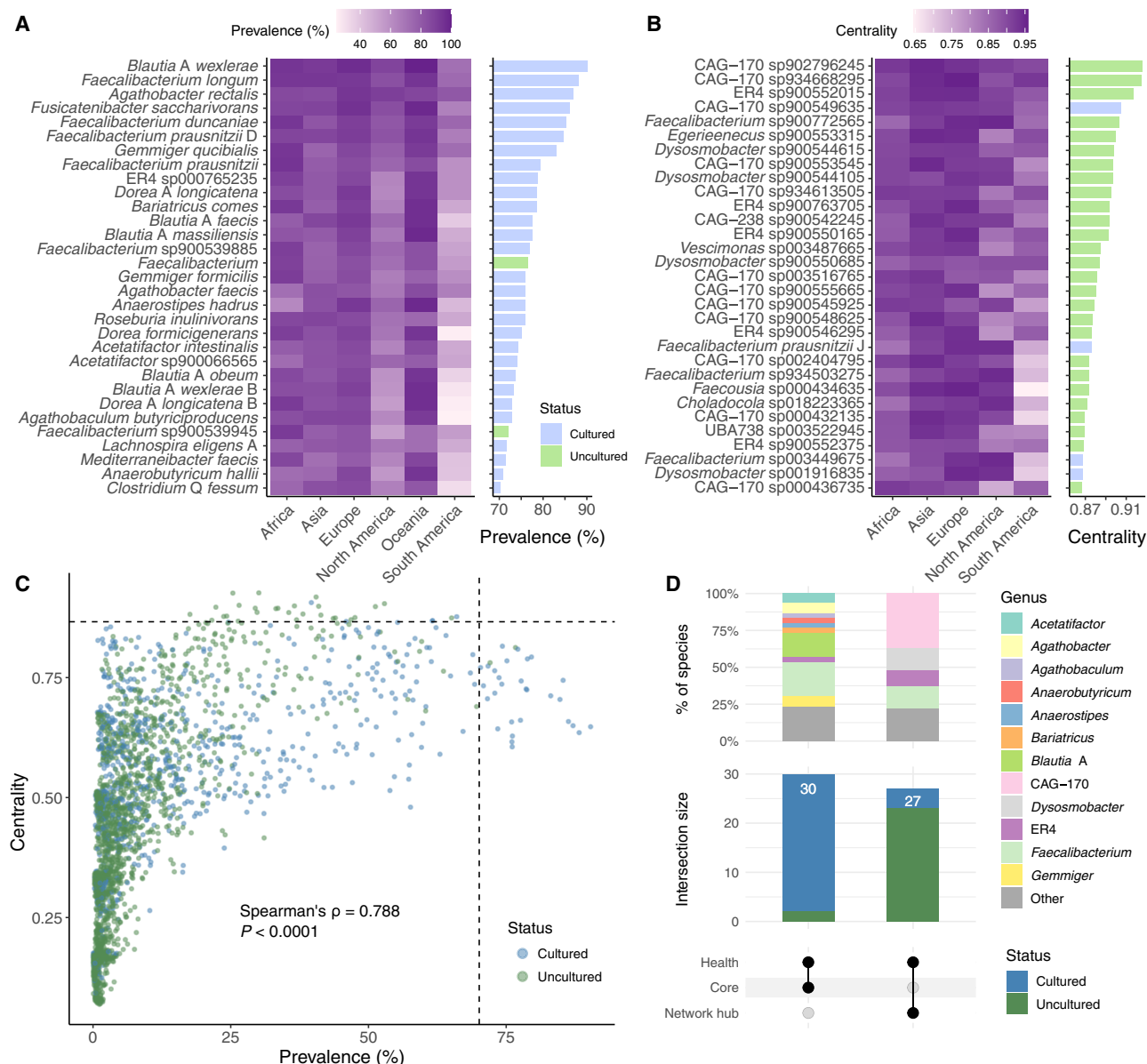


Figure 4. CAG-170 is the top network hub among healthy microbiomes worldwide

(A) Prevalence of the top 1% most common species (defined as “core”). The heatmap shows prevalence across continents, and the bar plot summarizes mean prevalence worldwide.

(B) Centrality scores of the top 1% of network hubs. The heatmap shows continent-specific values, and the bar plot shows the mean centrality score across continents.

(C) Distribution of mean prevalence and centrality scores for cultured and uncultured species. Dashed lines mark the thresholds we defined for core and hub species.

(D) Overlap among species classified as health-associated, core, and network hubs. Only species that overlapped in at least two groups are shown. Vertical bars indicate the number of species in each intersection, with stacked bars above showing their taxonomic affiliation (genus level).

the global healthy microbiome and that this signal is not strictly driven by a close association with other health- or disease-associated taxa.

Temporal stability and subspecies diversity of CAG-170

Given that cross-sectional case-control cohorts cannot capture microbiome dynamics over time, we further characterized the stability, abundance, and subspecies diversity of CAG-170

within a longitudinal cohort of IBD patients and controls.¹¹ This cohort was recruited as part of the second phase of the Human Microbiome Project (HMP2), characterizing a total of 132 Crohn’s disease, ulcerative colitis, and non-IBD individuals across a year, sampled at ~every 2 weeks. Each microbiome sample had an associated “dysbiosis” score, measuring the extent of deviation from a healthy gut microbiome composition inferred in the original study.¹¹

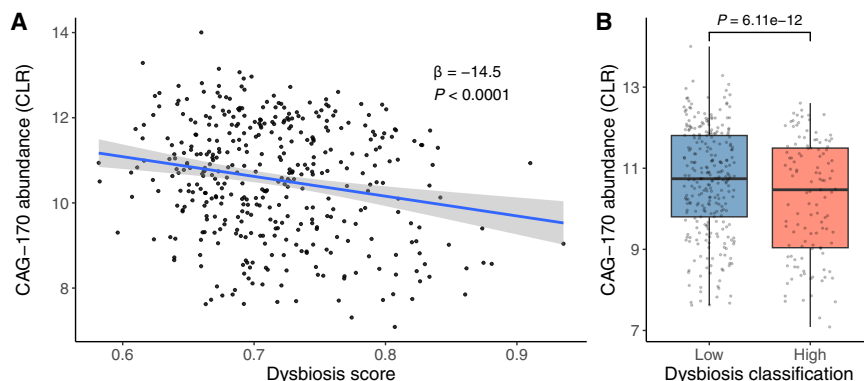


Figure 5. Association between CAG-170 abundance and lower dysbiosis

(A) Negative correlation between the abundance of CAG-170 (converted to centered log-ratio [CLR]) with the dysbiosis score estimated for each sample from the HMP2-IBD study. For visualization purposes only, samples with no CAG-170 species were excluded. The blue line depicts the fitted linear regression, with the shaded area representing the 95% confidence interval.

(B) Comparison of CAG-170 abundance after stratifying samples ($n = 1,118$) into “low” or “high” dysbiosis based on the median value. Within each boxplot, the center line represents the median. Whiskers extend to the furthest point within 1.5 times the IQR from the box. For visualization purposes only, samples with no CAG-170 species were excluded.

By reanalyzing this dataset with the UHGG catalog, we found that colonization of CAG-170 remained stable over a 1-year period across most subjects, with a median centered log-ratio (CLR) abundance difference between consecutive timepoints of 0.02 (Figure S10A) and of 0.13 when only considering subjects with CAG-170 detected in >10 timepoints (Figure S10B). Furthermore, the abundance of CAG-170 negatively correlated with the subject’s dysbiosis score (linear mixed-effects model, $p < 0.0001$; Figure 5A). This result was also consistent after classifying samples into either low or high dysbiosis, showing that the abundance of CAG-170 was significantly higher in low dysbiosis samples (linear mixed-effects model, $p = 6.11 \times 10^{-12}$; Figure 5B). Altogether, this supports the notion that CAG-170 is a stable member of the gut microbiome, and its abundance is negatively associated with the extent of microbial imbalance in the gut microbiome.

Next, to move beyond abundance- and species-level analyses, we assessed whether the subspecies genetic diversity of CAG-170 was associated with health and lower dysbiosis scores. We first generated pan-genomes of each of the 13 CAG-170 species, retrieving 20,263 genes from the Unified Human Gastrointestinal Protein (UHGP-90)²⁰ catalog. To assess the CAG-170 subspecies diversity, we screened the HMP2-IBD samples specifically for the presence of the accessory genes (that is, present in <90% of CAG-170 genomes). A total of 3,589 accessory genes were detected, of which 1,186 were found at >1% prevalence. By investigating the gene prevalence patterns among the HMP2-IBD samples with at least one CAG-170 accessory gene, we found 160 CAG-170 accessory genes negatively associated with dysbiosis score (FDR < 0.05; Table S6) and no genes with a positive association. The effect size of each significant accessory gene did not correlate with its prevalence in the CAG-170 clade (Pearson’s $r = -0.098$, $p = 0.221$; Figure S10C), indicating that the observed health signal was not driven by gene frequency among CAG-170 genomes. These health-associated genes covered various functional categories, including transport and nutrient acquisition (e.g., ATP-binding cassette transporters, ion pumps, and efflux systems), stress response and cellular maintenance (e.g., chaperones, DNA repair, and sporulation), and information processing (e.g., translation factors and transcriptional regulators). However, 87 candidate genes (54%) had no recognized function, representing a large uncharacterized diversity within

CAG-170 that remains unexplored. Overall, these results suggest that CAG-170 is an important uncultured taxon in gut health, with both its abundance and genetic diversity consistently linked to a low dysbiosis, health-associated microbiome.

CAG-170 shows enrichment of vitamin B12 genes and potential arginine auxotrophy

To obtain further biological insights into the CAG-170 genus, we performed a dedicated functional characterization of all high-quality (>90% completeness) genomes available from this clade ($n = 332$). We first investigated specialized primary and secondary metabolic pathways predicted with gutSMASH³⁴ and antiSMASH.³⁵ This revealed that three pathways in particular were the most prevalent among CAG-170 genomes (Figure S11A): (1) the Rnf complex representing a primary respiratory enzyme for anaerobic bacteria; (2) the porA pathway responsible for the conversion of R-pyruvate to R-acetate, a key short-chain fatty acid associated with gut health; and (3) secondary metabolic clusters involved in the production of β -lactones, metabolites with potential antimicrobial or signaling functions. Given the importance of carbohydrate metabolism in the gut microbiome, we expanded this analysis further and investigated which carbohydrate-active enzymes (CAZymes)³⁶ were most prevalent among CAG-170. A total of 43 CAZymes were detected among >50% of CAG-170 genomes, with 27 found in >90% of genomes (Figure S11B). These covered a wide range of functions, from cell-wall remodeling, glycogen metabolism, exopolysaccharide biosynthesis, and degradation of plant, chitin, and host-derived glycans, reflecting a wide carbohydrate metabolic capacity of CAG-170.

To further understand the unique functional repertoire of CAG-170, we further conducted a genomic analysis of the Oscillospiraceae family to pinpoint biological features that distinguish CAG-170 from closely related bacterial clades. To this end, we again focused solely on all high-quality and non-redundant genomes from the Oscillospiraceae family in the UHGG collection ($n = 4,025$ genomes from 152 species; Figure 6A). This dataset spanned 28 genera and was predominantly represented by genomes belonging to ER4 ($n = 1,210$), *Faecousia* ($n = 522$), and *Vescimonas* ($n = 445$). We applied a mixed-effects logistic regression model to compare the presence of KEGG Orthologs (KOs) in CAG-170 genomes versus all other Oscillospiraceae

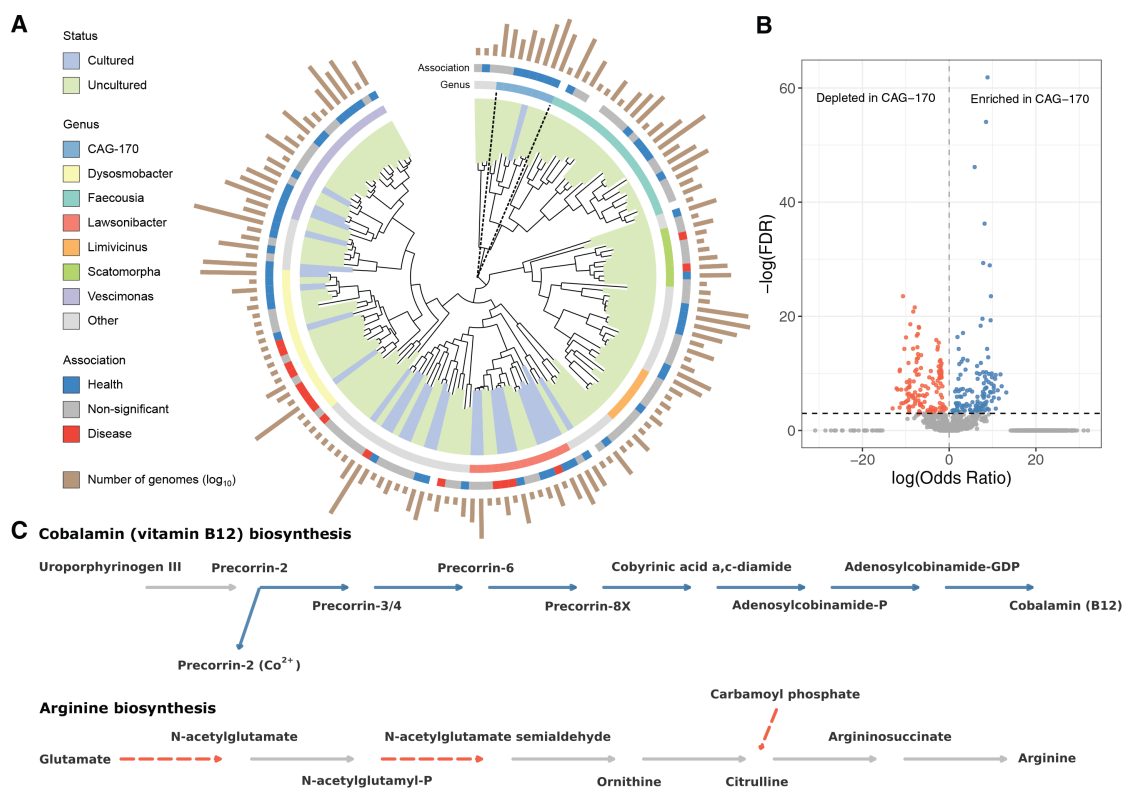


Figure 6. Phylogenetic and functional analysis of CAG-170

(A) Phylogenetic tree of 152 species from the Oscillospiraceae family, with clades and outer layers labeled by species culturing status, genus affiliation, and association with health state. The bar plot in the outermost layer denotes the $\log_{10}$ -transformed number of high-quality (>90% complete) genomes of the corresponding species present in the UHGG catalog. The clade belonging to the CAG-170 genus is highlighted with a dashed outline.

(B) Volcano plot showing the KEGG Orthologs (KOs) significantly enriched (positive effect size) or depleted (negative effect size) among CAG-170 genomes compared with other Oscillospiraceae genera. Adjusted p values (FDR) were derived from a mixed-effects logistic regression.

(C) Simplified diagram of the biosynthetic pathways for cobalamin (vitamin B12) and arginine production, focusing on reaction steps detected as significantly enriched (solid blue) or depleted (dashed red) in CAG-170.

genera. This analysis identified 135 KOs significantly enriched and 141 KOs significantly depleted in CAG-170 genomes (FDR < 0.05; Figure 6B; Table S7). Enriched genes spanned diverse metabolic pathways, most notably including 14 KOs covering the cobalamin (vitamin B12) biosynthesis pathway (Figure 6C). They also encompassed genes involved in methionine and serine biosynthesis (*metA*, *metY*, and *serA*), stress tolerance and DNA repair functions (*mutL*, *mutY*, *uvrD/pcrA*, and *dinG*), sporulation-related genes (*spoVFA*, *spoVFB*, and *spoVG*), and broad nutrient uptake systems (e.g., multiple sugar transporters *msmK/msmX* and *rbsB*). By contrast, depleted genes were linked to pro-inflammatory or virulence-associated functions, including lipopolysaccharide/rhamnose biosynthesis (*rfbB*, *rfbC*, *rfbD*, *rmlB*, *rmlC*, and *rmlD*), spore germination proteins (*cwlJ* and *sleB*), and metal stress regulators (*cadC/smtB*). Several vitamin and nucleotide biosynthesis genes were also missing, such as those in the thiamine (*thiD*, *thiE*, *thiM*, and *tenA*) and pantothenate (*panE*) pathways. Notably, some of the most strongly depleted genes among CAG-170 genomes are involved in key steps of the arginine biosynthesis pathway (*carA*, *carB*, *argC*, and *argJ*; Figure 6C), indicating likely arginine auxotrophy. Coincidentally, the only reported successful cultivation of one CAG-170 isolate involved arginine supplementation in

the culture medium³⁰—an approach not employed in earlier culturing attempts^{3,37}—suggesting it may be important for the *in vitro* growth of CAG-170.

Given the strong signal observed for vitamin B12 biosynthesis genes, we further examined which genetic traits were associated with their presence and function in CAG-170. Using the presence-absence patterns of each KO among the CAG-170 genomes, we calculated pairwise ϕ correlation coefficients between B12 biosynthesis KOs and all non-B12 KOs (Figure S11C). We found that the *hisC* gene in particular—encoding a histidinol-phosphate aminotransferase—showed the strongest correlation with B12 genes (Figure S11C). The co-occurrence between histidine and vitamin B12 biosynthesis genes may reflect a shared metabolic relationship, as vitamin B12 has been shown to bind its cobalt atom through a histidine residue,³⁸ potentially linking histidine metabolism to cofactor binding affinity.

CAG-170 exhibits cross-feeding potential with its co-colonizing species

Since our network analysis revealed that CAG-170 is highly interconnected with numerous other species in the gut microbiome, we further explored the potential biological basis of these

associations. We constructed genome-scale metabolic models for CAG-170 and its strongest co-colonizers (those within the top 1% of positive correlations), comparing the number and types of metabolites predicted from uptake and secretion fluxes. Notably, CAG-170 was predicted to secrete a significantly greater number of metabolites (Wilcoxon rank-sum test, $p < 0.0001$) than its co-colonizers, while exhibiting a significantly lower number of metabolites from uptake fluxes (Figure S12A). We identified 7 uptake and 28 secreted metabolites in particular as differentially present between CAG-170 and its co-colonizers (mixed-effects logistic regression, FDR < 0.05; Figure S12B). These metabolites included a mixture of amino acids, small organic acids, and cofactors, with two in particular—L-glutamine and L-threonine—showing preferential secretion by CAG-170 and preferential uptake by co-colonizers (Figure S12C).

As vitamin B12 biosynthesis was found to be a unique trait of CAG-170, we also examined further whether KOs associated with enzymes dependent on B12 as a cofactor were detected in either CAG-170 or its co-colonizers. We focused specifically on the *mut* gene (K01847) encoding a methylmalonyl-CoA mutase and the *metH* gene (K00548) coding for a methionine synthase, two enzymes known to be vitamin B12-dependent.³⁹ Interestingly, we found that both genes were significantly enriched (Fisher's exact test, $p < 0.0001$) among the co-colonizers. In fact, *mut* and *metH* were entirely absent from all CAG-170 genomes but present in 245 (12%) and 148 (7%) of co-colonizers' genomes, respectively. Collectively, these findings highlight CAG-170 as a key metabolic producer that may contribute to cross-feeding interactions within the human gut microbiome.

DISCUSSION

Here, we provide a large-scale, global analysis of the human gut uncultured microbiome across a broad spectrum of diseases, populations, and geographic regions. By integrating metagenomics with genome-based species profiling, machine learning, network analysis, and functional prediction, we reveal that uncultured bacterial species—particularly those in the CAG-170 genus—are widespread, clinically relevant, and carry unique ecological and functional signatures.

Despite representing only ~30% of species and ~12% of the total microbial abundance per sample, uncultured bacteria were found to provide discriminative signatures of microbial diversity and health status across multiple datasets. Among the uncultured bacteria identified, health-associated species were more commonly conserved across multiple diseases, whereas disease-associated species tended to be disease specific. A prior meta-analysis using 16S rRNA genotyping identified non-specific shared microbial responses across multiple conditions in both health and disease.⁶ Here, we identified a more limited overlap between diseased subjects, which could reflect inherent differences in analysis resolution between metagenomics and the 16S rRNA gene. Our study also further complements previous efforts investigating the unique properties of the healthy microbiome^{40,41} by focusing specifically on the insights obtained from uncultured gut bacteria.

Our most notable finding was the identification of the uncharacterized genus CAG-170 as a consistent health-associated taxon. Previous studies have linked CAG-170 to lower body

mass index⁴² and improved cardiometabolic health.⁴³ Here, we expand on this research by providing five key results: (1) CAG-170 was the uncultured genus most significantly overrepresented in healthy individuals across diverse disease conditions and geographic regions, (2) CAG-170 was the most interconnected genus in global healthy populations out of any other genus—cultured or uncultured, (3) CAG-170 was found to be a temporally stable member of the gut microbiome in a longitudinal cohort, (4) the abundance and subspecies diversity of CAG-170 negatively correlated with microbiome dysbiosis over time, and (5) CAG-170 species encode unique biological features that distinguish it from other closely related taxa.

Indeed, functional characterization of CAG-170 genomes revealed distinct biological attributes, including an enrichment of various genes related to vitamin B12 biosynthesis, when compared with other species from the Oscillospiraceae family. In the gut microbiome, vitamin B12 serves as a cofactor for microbial enzymes, leading to short-chain fatty acid production and amino acid metabolism.^{44–46} Therefore, a higher capacity for vitamin B12 biosynthesis may partly underlie the health-associated role of CAG-170. Furthermore, we also detected a depletion of pro-inflammatory genes and those involved in arginine biosynthesis among CAG-170 species, the latter of which could be related to the difficulty in culturing this genus. As a result, CAG-170 remains largely understudied. In fact, since nearly all available CAG-170 genomes are metagenome-assembled (only 1 isolate genome is currently available³⁰ while 1,046 CAG-170 MAGs are included in the UHGG catalog²⁰), 16S rRNA gene sequences for this group are also underrepresented in established databases such as SILVA.⁴⁷ This reflects a broader challenge in microbiome research: potentially relevant and ecologically important taxa remain invisible to traditional culturing techniques and certain marker gene surveys.

In conclusion, our findings highlight the uncultured gut microbiome as a fundamental and underappreciated component of human health. As microbiome research continues to evolve, future efforts should integrate metagenomics, culturing, and experimental validation to characterize these hidden yet biologically relevant species. Although it is unlikely that these species are truly “unculturable”—they are simply yet to be cultured—our study offers valuable insights for guiding future culturing strategies. This will not only improve our understanding of the role of the microbiome in health and disease but may also inform the development of next-generation microbial therapeutics that leverage the full diversity of the gut ecosystem.

Limitations of the study

While we analyzed a large and diverse range of diseases and populations, it is important to note some of the limitations of our study. First, the analysis of several diseases (especially of understudied conditions) may have been underpowered to detect subtle microbiome associations, particularly when analyzing low-abundant, uncultured species. Secondly, in our differential abundance statistical modeling, the covariates we could control for were limited by the available sample metadata. Additional factors such as diet and medication should be carefully considered in future, more targeted studies. Lastly, because our results are based on statistical associations and genomic predictions, they do not establish microbiome-disease causality

or direct ecological interactions. However, the lack of available isolates for these uncultured species currently limits further experimental work. Expanding cultivation efforts with targeted media supplementation, alongside continued development of genome-resolved metagenomics, will be essential for closing these knowledge gaps.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Alexandre Almeida (aa2369@cam.ac.uk).

Materials availability

This study did not generate new, unique reagents.

Data and code availability

- The human gut metagenomic samples used in this study are publicly available in the European Nucleotide Archive (see [Table S1](#) for all associated accession codes).
- The genome database used is available in the UHGG catalog v.1.0 (study accession, ENA: ERP116715).
- Abundance data estimated for the UHGG species and all metagenomic samples included here are available in figshare at <https://doi.org/10.6084/m9.figshare.29963597.v1>.
- The protein FASTA file of the CAG-170 accessory genes detected in the HMP2-IBD study can be accessed from <https://doi.org/10.6084/m9.figshare.29963588.v1>.
- Accession code of the human reference genome used for decontamination (GRCh38) is GenBank: GCA_000001405.15.
- Custom scripts and pipelines used in this work are publicly available in GitHub at <https://github.com/microfundiv-lab/SecretBugs>.

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AUTHOR CONTRIBUTIONS

A.C.d.S. curated the sample metadata, performed metagenomic analyses, and helped write the manuscript draft. J.L. performed the subspecies genetic analysis of CAG-170. Q.Y. assisted in the metagenomic analyses and data curation. E.M. co-supervised the work and contributed to the subspecies and statistical analyses of CAG-170. A.A. supervised the work, performed metagenomic analyses, and wrote the manuscript draft. All authors read, edited, and approved the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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- [METHOD DETAILS](#)
 - Metagenomic data collection and preprocessing
 - Generating a species-level genome database

- Species classification and cultured status
- Profiling species abundances
- Uncultured prevalence and diversity across health states
- Machine learning modelling for disease classification
- Differential species abundance analysis
- Meta-analysis across disease cohorts
- Network analysis in healthy populations
- Characterization of CAG-170 in a longitudinal cohort
- Genome functional annotation
- Oscillospiraceae phylogenetic and functional analysis
- Metabolic modelling of CAG-170 and its co-colonizers

● QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
European Nucleotide Archive	Yuan et al. ⁴⁸	https://doi.org/10.1093/nar/gkad1067
Human genome (GRCh38)	NCBI	GCA_000001405.15
UHGG / UHGP catalog	Almeida et al. ²⁰	http://ftp.ebi.ac.uk/pub/databases/metagenomics/mgnify_genomes/human-gut/v1.0/
CAG-170 accessory genes	This paper	https://doi.org/10.6084/m9.figshare.29963597.v1
Software and algorithms		
Fastq-dl (v.2.0.4)	GitHub	https://github.com/rpetit3/fastq-dl
curatedMetagenomicData	Pasolli et al. ⁴⁹	https://github.com/waldronlab/curatedMetagenomicData
GMrepo	Dai et al. ⁵⁰	https://gmrepo.humangut.info
TrimGalore (v.0.6.0)	Martin ⁵¹	https://doi.org/10.14806/ej.17.1.200
BWA-MEM (v.0.7.16a-r1181)	Li and Durbin ⁵²	https://github.com/lh3/bwa
CheckM (v.1.0.11)	Parks et al. ⁵³	https://doi.org/10.1101/gr.186072.114
GUNC (v.1.0.3)	Orakov et al. ⁵⁴	https://doi.org/10.1186/s13059-021-02393-0
Mash (v.2.3)	Ondov et al. ⁵⁵	https://doi.org/10.1186/s13059-016-0997-x
DNAdiff (v.1.3)	Kurtz et al. ⁵⁶	https://doi.org/10.1186/gb-2004-5-2-r12
GTDB-Tk (v.2.3.2)	Chaumeil et al. ⁵⁷	https://doi.org/10.1093/bioinformatics/btz848
SAMtools (v.1.9)	Li et al. ⁵⁸	https://doi.org/10.1093/bioinformatics/btp352
ConQuR (v.1.2.0)	Ling et al. ⁵⁹	https://doi.org/10.1038/s41467-022-33071-9
Mikropml	Topcuoglu et al. ⁶⁰	https://doi.org/10.21105/joss.03073
ALDEX2 (v.1.32.0)	Fernandes et al. ²⁸	https://doi.org/10.1371/journal.pone.0067019
MaAsLin2 (v.1.14.1)	Mallick et al. ²⁹	https://doi.org/10.1371/journal.pcbi.1009442
FastSpar (v.1.0)	Watts et al. ⁶¹	https://doi.org/10.1093/bioinformatics/bty734
Igraph	Csardi and Nepusz ⁶²	https://doi.org/10.5281/zenodo.17470081
Prokka (v.1.14.16)	Seemann ⁶³	https://doi.org/10.1093/bioinformatics/btu153
KOFams (release 2021–11)	Aramaki et al. ⁶⁴	https://doi.org/10.1093/bioinformatics/btz859
FastTree (v.2.1.11)	Price et al. ⁶⁵	https://doi.org/10.1093/molbev/msp077
MEGAHIT (v.1.2.9)	Li et al. ⁶⁶	https://doi.org/10.1093/bioinformatics/btv033
Prodigal (v.2.6.3)	Hyatt et al. ⁶⁷	https://doi.org/10.1186/1471-2105-11-119
DIAMOND (v.2.1.8)	Buchfink et al. ⁶⁸	https://doi.org/10.1038/nmeth.3176
gutSMASH (v.1.0)	Andreu et al. ³⁴	https://doi.org/10.1038/s41587-023-01675-1
antiSMASH (v.6.0.1.)	Blin et al. ³⁵	https://doi.org/10.1093/nar/gkab335
dbCAN (v.2.0.11)	Zhang et al. ⁶⁹	https://doi.org/10.1093/nar/gky418
CarveMe (v.1.5.2)	Machado et al. ⁷⁰	https://doi.org/10.1093/nar/gky537
COBRAPy (v.0.29)	Ebrahim et al. ⁷¹	https://doi.org/10.1186/1752-0509-7-74
metaMap	Yin et al. ²¹	https://github.com/alexmsalmeida/metamap
ML-Microbiome	Yin et al. ²¹	https://github.com/alexmsalmeida/ml-microbiome

METHOD DETAILS

Metagenomic data collection and preprocessing

In this study, we collected 11,115 human gut metagenomic samples from the European Nucleotide Archive (ENA),⁴⁸ spanning 62 studies and 39 countries. This dataset included 8,672 metagenomes from 36 case-control studies of 13 noncommunicable diseases, as well as an additional 2,443 samples from 26 studies composed exclusively of healthy individuals (Table S1). Sample metadata were extracted and manually curated by reviewing the associated publications and supplemented with information from the curatedMetagenomicData⁴⁹ and GMrepo⁵⁰ databases. Samples were selected based on the following criteria: (1) availability of metadata on health status, age group, and country of origin; (2) absence of diagnosed acute infections; and (3) no reported antibiotic use within

the previous month. Raw metagenomic datasets were downloaded from ENA using fastq-dl (v.2.0.4; <https://github.com/rpetit3/fastq-dl>), followed by quality control using TrimGalore (v.0.6.0).⁵¹ Human DNA contamination was removed by aligning reads to the human reference genome (GRCh38) using BWA-MEM (v.0.7.16a-r1181).⁵² Samples with fewer than 500,000 paired-end metagenomic reads after filtering were excluded from further analysis.

Generating a species-level genome database

To generate a custom database of genomes for metagenomic read mapping, we gathered all species representatives genomes from the Unified Human Gastrointestinal Genome (UHGG) catalog (v.1)²⁰ and filtered those matching all of the following criteria, as previously described²¹: (1) singletons (that is, the respective species includes only one genome); (2) <90% genome completeness based on CheckM (v.1.0.11)⁵³; and (3) classified by GUNC (v.1.0.3)⁵⁴ as chimaeric ('clade_separation_score' >0.45, 'contamination_portion' >0.05 and 'reference_representation_score' >0.5). This removed 32 species, resulting in a total of 4,612 species representatives. The final curated database was indexed using BWA (v.0.7.16a-r1181, 'bwa index') for subsequent read mapping.

Species classification and cultured status

Species classification based on their cultured status was determined according to two main criteria: (1) the genome type of the species representative (isolate or MAG); and (2) similarity to isolate genomes present in the NCBI RefSeq database (release 219). For this purpose, each genome was first screened against the RefSeq database using Mash (v.2.3)⁵⁵ to identify its most similar relative (lowest Mash distance). Subsequently, DNAdiff (v.1.3)⁵⁶ was used to compute whole-genome average nucleotide identities (ANI) between each pair. A species was considered 'cultured' if it was represented by an isolate genome or had >95% ANI over a minimum 30% alignment fraction to its closest RefSeq isolate. This classification was further refined through manual curation informed by literature searches, for example, when newly published UHGG species isolates were not represented in RefSeq. In addition, to account for taxonomy updates since the initial release of the UHGG catalog, species representatives were reclassified with GTDB-Tk (v.2.3.2) (database release 214)⁵⁷ using the 'classify_wf' with default parameters.

Profiling species abundances

Each quality-filtered metagenomic sample was mapped and quantified against our custom UHGG catalog, using a previously validated read mapping-based pipeline.^{17,21} Briefly, metagenomic reads were first aligned using BWA-MEM and processed with SAMtools (v1.9).⁵⁸ For each species in the reference database, we calculated the following metrics per sample: breadth of coverage, depth of coverage, total read counts and counts of uniquely mapped reads. To account for differences in sequencing depth between samples, we computed the expected breadth of coverage per species per sample using an established formula.^{10,21} A species was considered present in a sample if it met both of the following criteria: (1) a ratio of observed to expected breadth of coverage exceeding 30% (considering that genomes were clustered at a species level using a 30% aligned fraction); and (2) an observed breadth of coverage >5% (to account for MAGs being up to 5% contaminated). These coverage thresholds and presence/absence criteria were previously validated using both mock community and synthetic datasets.²¹ Species not meeting these thresholds were assigned a read count of zero.

To correct the filtered read counts for study-specific batch effects, we applied a conditional quantile normalization algorithm implemented in ConQuR (v.1.2.0).⁵⁹ The sample by species matrix of read counts was used as input and each of the 62 studies was evaluated as a potential reference for batch correction. Based on PERMANOVA results, study SRP188507 was selected as the final reference, as the corrected read counts yielded the lowest study effect size ($R^2 = 0.088$). To minimize technical artifacts and false positives, we enforced a read count of zero after batch correction for species that did not meet our coverage thresholds described above.

Uncultured prevalence and diversity across health states

We assessed the association between the total prevalence of uncultured species or alpha diversity with disease status by comparing the patterns observed between case and control samples. For each sample, uncultured prevalence was measured as the proportion of uncultured species among all detected species. Alpha diversity was calculated from the Shannon index and estimated separately after partitioning the read count matrix into cultured and uncultured species. Thereafter, we used a linear mixed-effects model to evaluate the association between the response variable (uncultured prevalence or alpha diversity) with disease status, while controlling for sequencing depth and study source. If applicable, the subject ID was also used as a random intercept to account for repeated measures. The *P* value corresponding to the effect of disease status was extracted from the model summary and corrected for multiple testing across diseases using the Benjamini-Hochberg method. Statistical significance of the alpha diversity results was further confirmed using relative abundances normalized by genome length, and with or without rescaling each species fraction (cultured or uncultured) to 100% (Table S2).

Machine learning modelling for disease classification

We developed supervised machine learning classifiers to assess the predictive potential of the microbiome for disease classification. For each disease, we conducted three types of analyses: (1) pooled analysis of studies from the same disease, (2) cross-study validation within the same disease, and (3) pairwise cross-disease validation (i.e., training the model with one disease and testing it on another). All machine learning analyses were performed using a custom workflow (<https://github.com/alexmsalmeida/ml-microbiome>) derived from the Mikropml R package.⁶⁰ Batch-corrected abundances of filtered microbiome species were transformed

using the centred log-ratio (CLR) and used as input features for the machine learning models. Prior to modelling, features were filtered to exclude those with <1% prevalence across all samples or with zero variance. Model training and hyperparameter tuning were performed on 80% of the data using 5-fold cross-validation, while the remaining 20% was held out for testing using the best hyperparameters. This procedure was repeated 20 times with different random seeds to ensure robustness. For the pooled analysis, 'subject ID' was used as a grouping factor to ensure that samples from the same individual (e.g., from repeated measures across time) were assigned exclusively to either the training or testing set. In the cross-study validation within the same disease, 'study' was used as the grouping factor instead of 'subject ID'. Model performance was assessed using the Area Under the Receiver Operating Characteristic Curve (AUROC). Three supervised learning algorithms — ridge regression, random forest, and gradient boosting — were evaluated with the pooled dataset (that is, after aggregating all samples across studies from the same disease). AUROC values across seeds and diseases were compared between models using a Kruskal-Wallis and a two-sided Wilcoxon rank-sum test. *P* values were corrected for multiple testing using the Bonferroni–Holm method. Ridge regression was selected for all subsequent analyses based on the superior performance observed (adjusted *P* < 0.05). The performance of ridge regression models using uncultured species only, cultured species only, and both combined were compared per disease with a two-sided Wilcoxon rank-sum test. *P* values were corrected for multiple disease comparisons using the Benjamini–Hochberg method.

Differential species abundance analysis

To identify microbial species associated with health and disease in case–control cohorts, we employed two complementary generalized linear modelling approaches: ALDEx2 (v1.32.0)²⁸ and MaAsLin2 (v1.14.1).²⁹ We performed the analysis separately for each disease, with health status, age group, continent, sequencing depth, and study source included as covariates in the models. For ALDEx2, we applied the 'aldex.clr' function to perform a centred log-ratio (CLR) transformation on the abundance data (read counts), followed by 'aldex.glm' to model differential abundance between health and disease. For MaAsLin2, read counts were normalized by genome length and sequencing depth using total sum scaling, then log-transformed prior to modelling. In both methods, species with a prevalence below 1% across samples were excluded to reduce sparsity. Multiple testing correction across species was performed using the Benjamini–Hochberg method separately for each tool, and microbial species were considered significant only if they passed a false discovery rate (FDR) threshold <5% in both ALDEx2 and MaAsLin2.

Meta-analysis across disease cohorts

We summarized species-level results per disease by first averaging the effect sizes obtained with ALDEx2 and MAAsLin2. Thereafter, effect sizes were summed across diseases and normalized by the number of diseases in which the species was found non-significant, resulting in a combined effect size that prioritized species showing consistent significance across multiple diseases. To identify the most promising uncultured taxa linked to health, we further calculated an uncultured health score for each bacterial genus present in the UHGG catalog. This metric was based on the product of three factors: (i) the total number of health-associated species in the genus (log₁₀-transformed); (ii) the combined effect sizes normalized by the number of species in the genus; and (iii) the proportion of uncultured species. In both (ii) and (iii), the denominator included only species within the genus that had a minimum prevalence of 1% in at least one case-control cohort. We summarized the results at the genus level, as this approach enabled us to evaluate the degree of novelty within each taxon by incorporating the number of uncultured species in our calculation. Moreover, because uncultured genomes are represented by MAGs, which are inherently lower in quality compared to isolate genomes, prioritizing taxa where multiple species and genomes have been identified independently increased overall confidence in the robustness of the results.

To validate our approach using an alternative methodology, we performed an independent meta-analysis using the classical Stouffer's method. A limitation of this approach is that it cannot accommodate the initial FDR thresholding applied when intersecting the ALDEx2 and MaAsLin2 results. Therefore, for this validation we relied solely on ALDEx2, which is recognized as the most conservative differential abundance method.⁷² Using this approach, the ALDEx2 effect sizes obtained per species were weighted across diseases by the square root of the corresponding sample size, and Z-scores were derived from the reported *P* values. The weighted Z-scores were then aggregated to obtain a meta-analysis Z-score, which was converted into a combined *P* value. Multiple testing correction using the Benjamini–Hochberg method was performed across species based on this combined *P* value, and statistical significance was assessed at an FDR threshold <5%.

Network analysis in healthy populations

To conduct a dedicated network analysis of the healthy human microbiome, we compiled all control samples from the case-control studies included in this work (*n* = 3,614), along with an additional dataset of 2,443 metagenomes from healthy subjects. We excluded all samples from infants and children under 13 years of age. Microbial correlation networks were constructed using FastSpar (v.1.0),⁶¹ a C++ implementation of SparCC,³¹ based on species with a prevalence greater than 1%. Abundance values used in FastSpar were derived from the batch-corrected read counts. Co-abundance networks were generated separately for samples from five continents (Asia, Africa, Europe, North America and South America), with samples from Oceania excluded due to insufficient sample size. Thereafter, within each network we calculated exact *P* values from 1,000 bootstrap replicates, applying a Benjamini–Hochberg multiple testing correction to retain correlations with FDR <5%. SparCC correlation values were transformed into distances (calculated as 1 – |correlation|), and for each species, we computed three network centrality measures — betweenness, closeness, and degree — using the igraph R package.⁶² Betweenness centrality reflects how often a species lies on the shortest paths between others, closeness centrality indicates how close a species is to all others in the network, and degree centrality represents the number

of direct connections a species has. Previous studies have shown that species with high closeness, high degree and low betweenness are more likely to be keystone.^{32,33} Therefore, we rescaled each metric with betweenness values inverted (multiplied by -1), and averaged across continents to derive a single score, referred to as 'centrality score'. Species within the top 1% centrality score were classified as network hubs. Results were repeated by further filtering the initial co-abundance networks to only retain edges with $|\text{correlation}| > 0.1$ and with $|\text{correlation}| > 0.2$.

Characterization of CAG-170 in a longitudinal cohort

We analysed 1,118 metagenomic samples from a longitudinal IBD cohort of the Human Microbiome Project (HMP2-IBD)¹¹ to further investigate the stability and subspecies genetic diversity of the CAG-170 taxon over time in health and disease. This cohort included individuals diagnosed with Crohn's disease, ulcerative colitis, and non-IBD controls. Participants were originally followed for one year, with stool samples collected approximately every two weeks. Each microbiome sample was associated with a 'dysbiosis' score, quantifying the deviation from a healthy gut microbial composition as defined in the original HMP2 study.

We first evaluated the prevalence and temporal stability of CAG-170 over the one-year period. For each subject, stability was quantified by calculating the centred log-ratio (CLR) abundance difference of CAG-170 between consecutive timepoints. Given a low prevalence of CAG-170 in some subjects could result in a misleading signal of stability, we performed this analysis both for subjects that harboured CAG-170 in at least one timepoint and in those with CAG-170 detected in >10 timepoints.

To evaluate the association between CAG-170 abundance and 'dysbiosis', we applied a linear mixed-effects model. Specifically, we modelled the CAG-170 genus-level abundance as a function of the dysbiosis score, with subject ID as a random effect to account for repeated measurements within subjects. The significance of the association between dysbiosis and CAG-170 abundance was assessed using an ANOVA test on the fitted linear mixed-effects model. The analysis was then repeated by partitioning dysbiosis scores into 'high' or 'low' based on the median value.

We further investigated the subspecies genetic diversity of CAG-170 within this cohort. First, we extracted all accessory genes from the Unified Human Gastrointestinal Protein (UHGP-90-HQ)²⁰ catalog exclusive to each of the 13 CAG-170 species, with accessory genes being defined as those present in fewer than 90% of conspecific genomes. For this purpose, only genomes with $>90\%$ completeness and $<5\%$ contamination, as defined in the UHGG database, were considered. First, metagenome assemblies were generated for all 1,118 HMP2-IBD metagenomic samples using MEGAHIT (v1.2.9)⁶⁶ with the '-min-contig-len 500' option. Thereafter, protein-coding sequences were predicted from these assemblies using Prodigal (v2.6.3)⁶⁷ with the '-p meta' parameter. To profile the gene presence across samples, we used DIAMOND (v2.1.8)⁶⁸ in 'blastp' mode to align all extracted CAG-170 accessory genes against the predicted protein sequences from the HMP2-IBD assemblies. We used an assembly- rather than read-based approach, as this enabled us to leverage the full length of the predicted gene to improve gene detection accuracy. Alignments were filtered using a threshold of $\geq 90\%$ amino acid identity and $\geq 80\%$ coverage of the shorter sequence. This resulted in a binary presence/absence matrix of accessory genes across all metagenomic samples. We kept only accessory genes that appeared in more than 1% of HMP2-IBD samples. To minimize the effect of CAG-170 species abundance in gene detection and downstream statistical analysis, we focused only on accessory genes (that is, found in $<90\%$ of conspecific genomes) and retained only HMP2-IBD samples in which we detected at least one accessory gene.

To assess the association between gene presence and dysbiosis, a mixed-effects logistic regression model was fitted, with subject ID as a random effect. *P* values were adjusted for multiple testing using the Benjamini-Hochberg method, and genes with FDR $<5\%$ were considered statistically significant. In addition, to account for potential detection biases due to gene length differences, we confirmed that the genes inferred to be significantly associated with health did not differ in length from the non-significant genes using a two-sided Wilcoxon rank-sum test. Functional annotations, including eggNOG⁷³ and Clusters of Orthologous Groups (COG)⁷⁴ categories for the candidate accessory genes were obtained from the original UHGP catalog.

Genome functional annotation

We performed a functional prediction analysis of CAG-170, the Oscillospiraceae family and of CAG-170 co-colonizers using only high-quality ($>90\%$ complete) and non-redundant (99.9% average nucleotide identity) genomes. Protein-coding sequences were first extracted from the genome assemblies using Prokka (v1.14.16).⁶³ Each genome was then functionally characterized using KO-Fams (release 2021-11)⁶⁴ with default parameters. For the 332 high-quality CAG-170 genomes, the Prokka annotations were further used to estimate both specialized primary and secondary metabolic pathways using gutSMASH (v.1.0)³⁴ and antiSMASH (v.6.0.1)³⁵ using default parameters. Carbohydrate utilization enzymes (CAZymes) were also predicted using dbCAN (v.2.0.11)⁶⁹ using default parameters and further classified into broad functional classes through manual curation.

Oscillospiraceae phylogenetic and functional analysis

A phylogenetic and functional analysis of the Oscillospiraceae family was performed to identify features specifically associated with the genus CAG-170. First, a family-level phylogenetic tree was constructed using all high-quality species representative genomes of Oscillospiraceae from the UHGG collection ($n = 152$ species). The tree was generated with FastTree (v.2.1.11)⁶⁵ based on the alignment of bacterial marker genes obtained using GTDB-Tk (v.2.3.2, database release 214).⁵⁷ Next, from the predicted KEGG Ortholog (KO) annotations of all Oscillospiraceae genomes ($n = 4,025$ genomes), we investigated which functional traits were associated with the CAG-170 genus, compared to the rest of the Oscillospiraceae family, using a mixed-effects logistic regression analysis. Only KOs found at a prevalence $>1\%$ were included. In fitting the model, the response variable was defined as the presence or absence of the

KO in each genome, and the predictor variable was genus membership (CAG-170 vs. other Oscillospiraceae). We included as a random effect the species affiliation to account for the fact there were multiple genomes from the same species under analysis. Species assignment was originally determined in the UHGG catalog²⁰ by (i) clustering all genomes at a 95% average nucleotide identity over a minimum 30% alignment fraction; and (ii) performing taxonomic assignment of the representative, highest quality genome within each cluster using GTDB-Tk.⁵⁷ *P* values from all models were corrected for multiple testing and KOs with an FDR <5% were considered statistically significant.

Metabolic modelling of CAG-170 and its co-colonizers

To assess the metabolic relationship between CAG-170 and its strongest co-colonizers, we generated genome-scale metabolic models for all high-quality CAG-170 genomes ($n = 332$) and for genomes belonging to species in the top 1% of positive correlations to CAG-170 ($n = 2,009$) identified in our network analysis. Metabolic models were generated from the Prokka annotations using CarveMe (v.1.5.2)⁷⁰ with default parameters and gap-filled with a gut-specific rich media (M3).⁷⁵ From the reconstructed models, we used a flux balance analysis using the COBRApy (v.0.29)⁷¹ Python package to estimate the number and types of metabolites predicted from secreted or uptake fluxes. Metabolites extracted were further analysed using a mixed-effects logistic regression model to compare CAG-170 with its co-colonizers. The metabolite presence/absence pattern was used as the response variable, and whether the genome was CAG-170 or its co-colonizer was set as the predictor variable. As above, species affiliation was included as a random effect. *P* values across all metabolites were corrected for multiple testing and those with an FDR <5% were considered statistically significant.

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were performed in R. Details of statistical tests used, sample size and meaning of error bars is reported in the figure legends and the [results](#) section. *P* values for all tests were corrected for multiple testing and statistical significance was determined based on an adjusted $P < 0.05$. As stated above, metagenomic samples not matching the following criteria were excluded: (1) availability of metadata on health status, age group, and country of origin; (2) absence of diagnosed acute infections; and (3) no reported antibiotic use within the previous month. Only high-quality (>90% completeness) genomes were used for functional prediction analysis and subsequent statistical tests.

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Supplemental information

**Meta-analysis of the uncultured gut microbiome
across 11,115 global metagenomes reveals
a candidate signature of health**

Ana C. da Silva, Jacob Lapkin, Qi Yin, Efrat Muller, and Alexandre Almeida

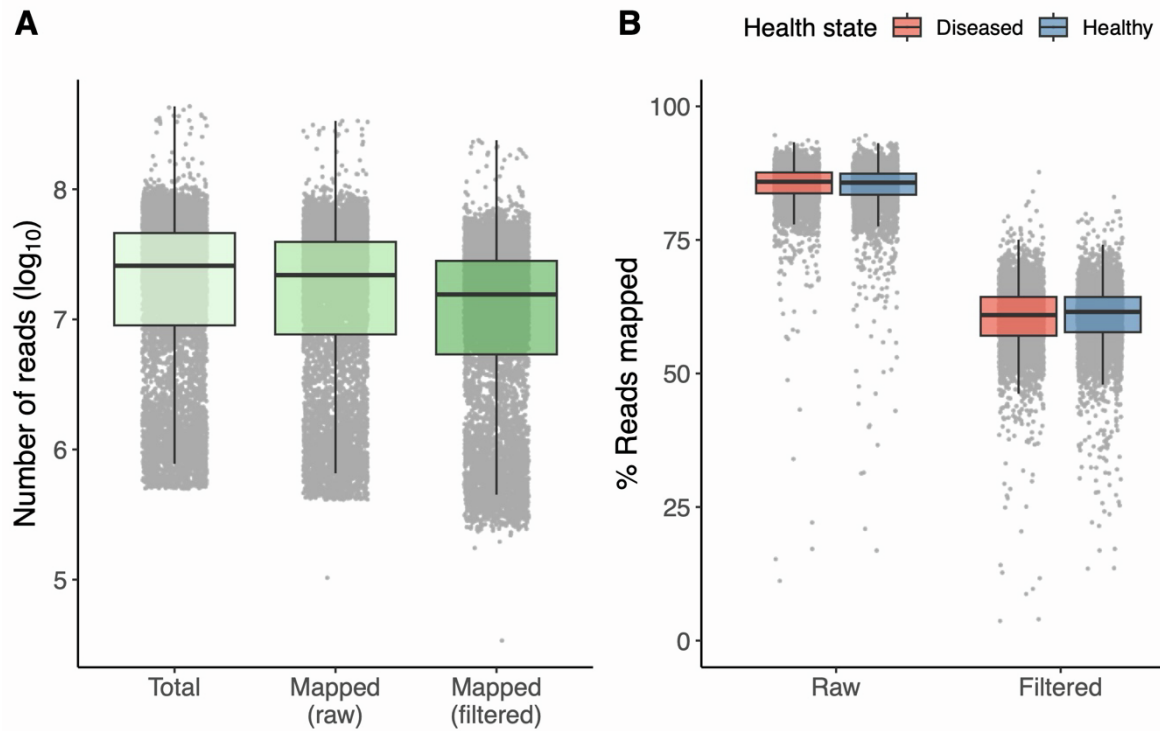


Figure S1. Distribution of sequencing depth and mapping rates. Related to STAR Methods.

(A) Distribution of the total number of reads, number of reads mapped (raw) and number of reads mapped after filtering among the case-control metagenomic samples ($n = 8,672$).

(B) Proportion of reads mapped across healthy and diseased samples without applying any mapping filters (“Raw”) and after applying filters based on observed and expected breadth of coverage (“Filtered”). The centre line within the box represents the median score. Whiskers are shown extending to the furthest point within 1.5 times the IQR from the box (disease samples, $n = 4314$; healthy samples, $n = 4358$).

abundance was inferred by dividing the number of reads mapped to uncultured species out of all reads mapped to the UHGG catalog. Vertical dashed lines represent the median values. *P* values were derived from a linear mixed-effect model and corrected for multiple testing using the Benjamini-Hochberg method. *FDR < 0.05; **FDR < 0.01; ***FDR < 0.001; N.S. = Non-significant.

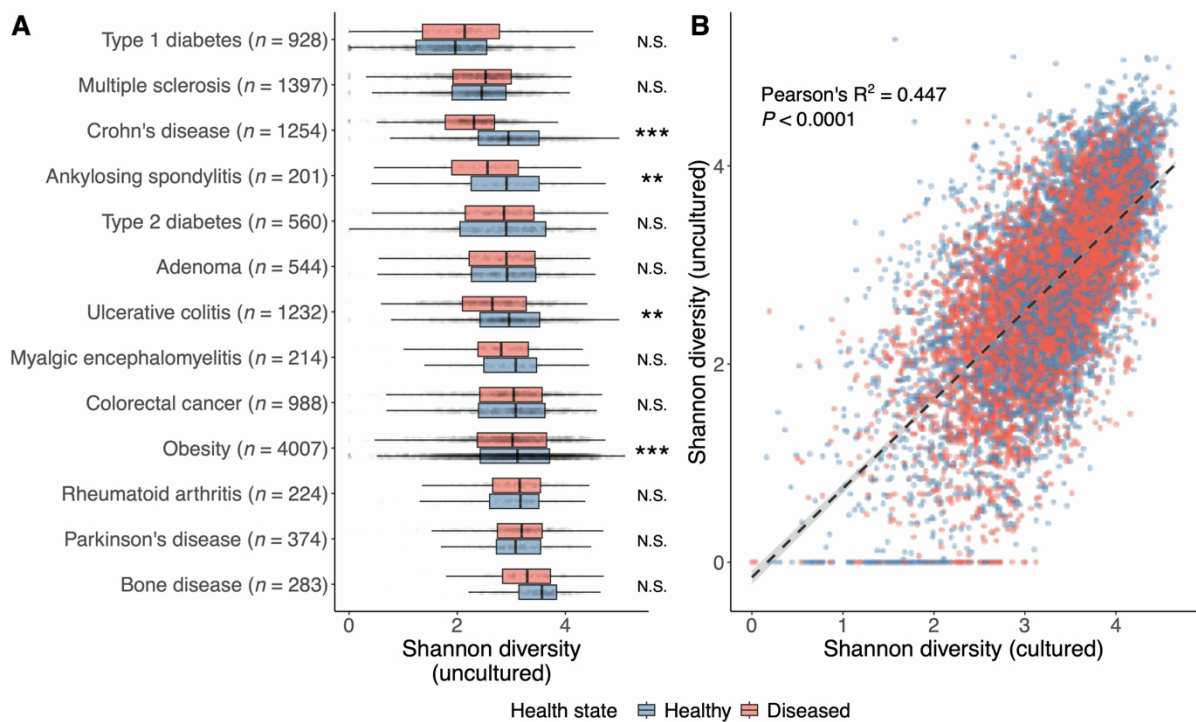


Figure S3. Alpha diversity estimates in health and disease. Related to STAR Methods.

(A) Distribution of alpha diversity values (Shannon index) across disease conditions, estimated using read counts of uncultured species only. P values were derived from a linear mixed-effects model with Shannon diversity as the response, and health state, study source, and sequencing depth (log-transformed) as fixed effects. The subject ID was used as a random intercept if there was more than one sample per subject. **adjusted $P < 0.01$; ***adjusted $P < 0.001$; N.S. = Non-significant. The centre line within the box represents the median score. Whiskers are shown extending to the furthest point within 1.5 times the IQR from the box. Total number of samples (n) per disease are indicated in parenthesis.

(B) Correlation between the Shannon alpha diversity estimated using either cultured or uncultured species. Samples are coloured by health state.

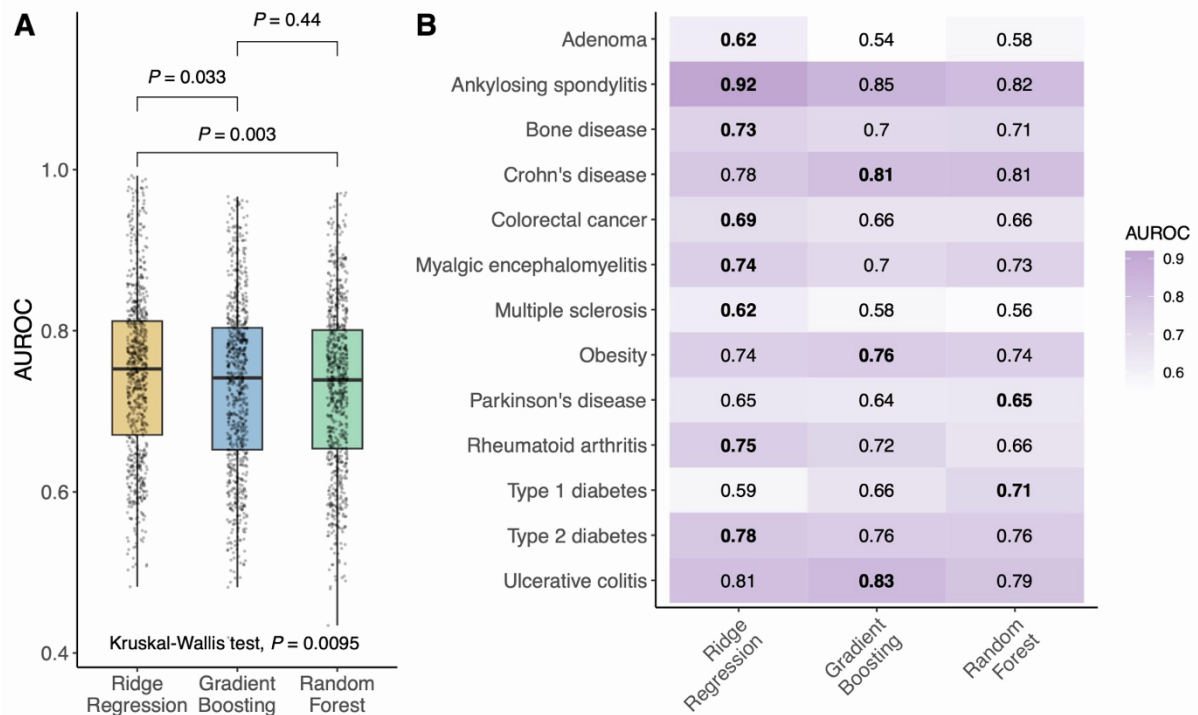


Figure S4. Evaluating the best supervised machine learning model. Related to STAR Methods.

(A) Model performance assessed by the Area Under the Receiver Operating Characteristic Curve (AUROC) for the three supervised machine learning methods tested. The centre line within the box represents the median score ($n = 780$, corresponding to 20 seeds per model per dataset — cultured, uncultured and both — across 13 diseases). Whiskers are shown extending to the furthest point within 1.5 times the IQR from the box. P values were derived from a two-sided Wilcoxon rank-sum test and corrected for multiple testing using the Bonferroni–Holm method.

(B) Performance of the models when using the uncultured microbiome alone, partitioned by disease. Values correspond to the median value across 20 seeds. The best performing model per disease is indicated in bold.

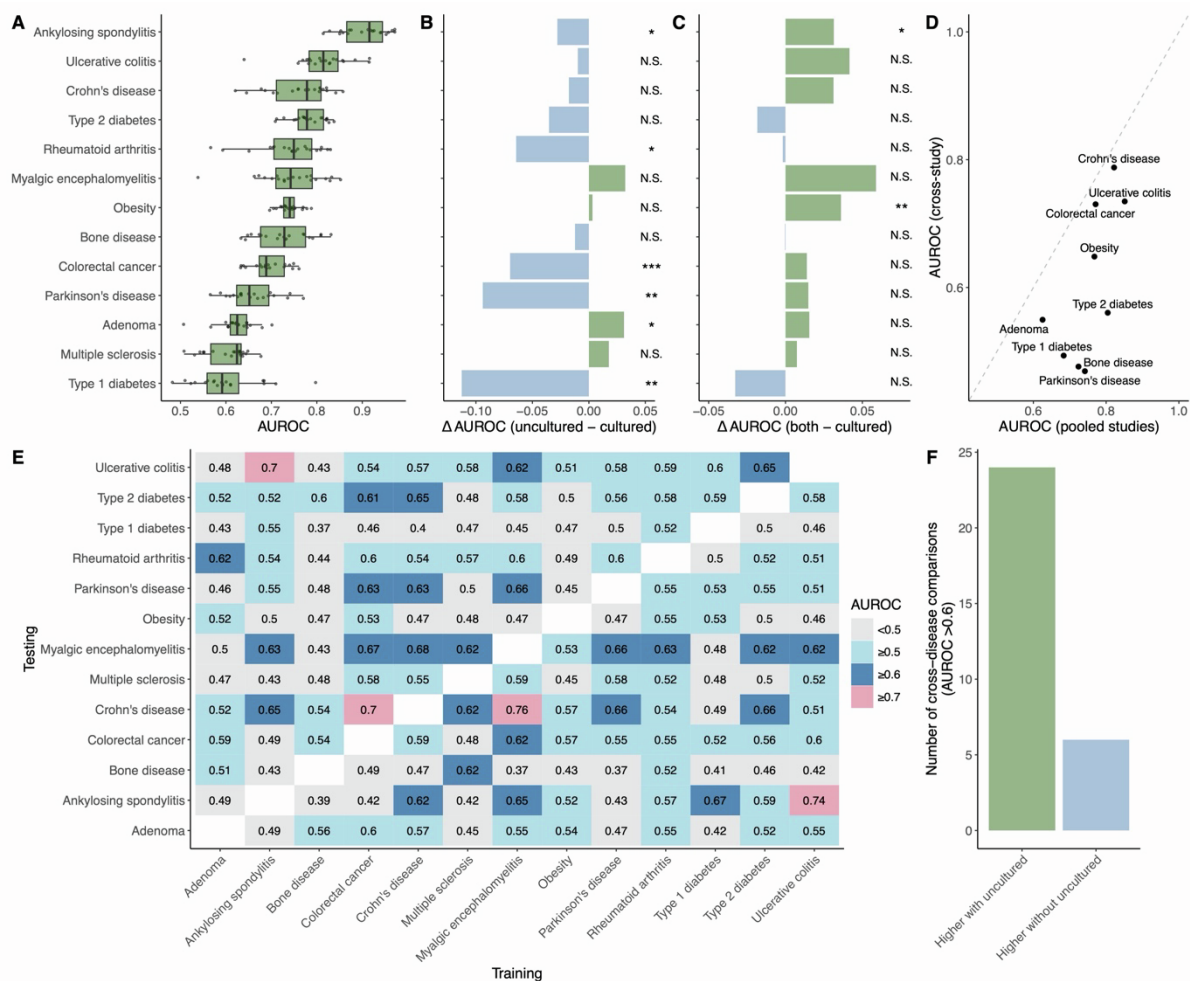


Figure S5. Performance of machine learning models for disease classification. Related to STAR Methods.

(A) Performance of the machine learning models based on the Area Under the Receiver Operating Characteristic Curve (AUROC) using only uncultured species as features. The centre line within the box represents the median score ($n = 20$ seeds per model). Whiskers are shown extending to the furthest point within 1.5 times the IQR from the box.

(B) Differences in performance between models using only uncultured or cultured species, measured as the difference in median AUROC across 20 seeds. Bars with positive values indicate that models based on uncultured species outperformed those based on cultured species only, and vice-versa. P values were derived from a two-sided Wilcoxon rank-sum test and

corrected for multiple testing using the Benjamini-Hochberg method. *FDR < 0.05; **FDR < 0.01; ***FDR < 0.001; N.S. = Non-significant.

(C) Performance differences between models with both uncultured and cultured species compared to models with cultured species only, measured as the difference in median AUROC across 20 seeds. Bars with positive values indicate the addition of uncultured species increased model performance. *P* values were derived from a two-sided Wilcoxon rank-sum test and corrected for multiple testing using the Benjamini-Hochberg method. *FDR < 0.05; **FDR < 0.01; N.S. = Non-significant.

(D) Comparison of model performance (median AUROC) when pooling studies from the same disease versus a cross-study validation (training on one set of studies and testing on others). Diagonal dashed line represents equal performance between both approaches. Models were built using both cultured and uncultured species.

(E) Model performance of a cross-disease validation, where models are trained on one disease and tested on another. AUROC values shown are medians across 20 random seeds. Models were built using both cultured and uncultured species.

(F) Number of cross-disease comparisons with an AUROC > 0.6 ($n = 30$) in which performance was significantly higher (two-sided Wilcoxon rank-sum test, FDR < 0.05) in models that included both cultured and uncultured species ('higher with uncultured') or in models that only used cultured species ('higher without uncultured').

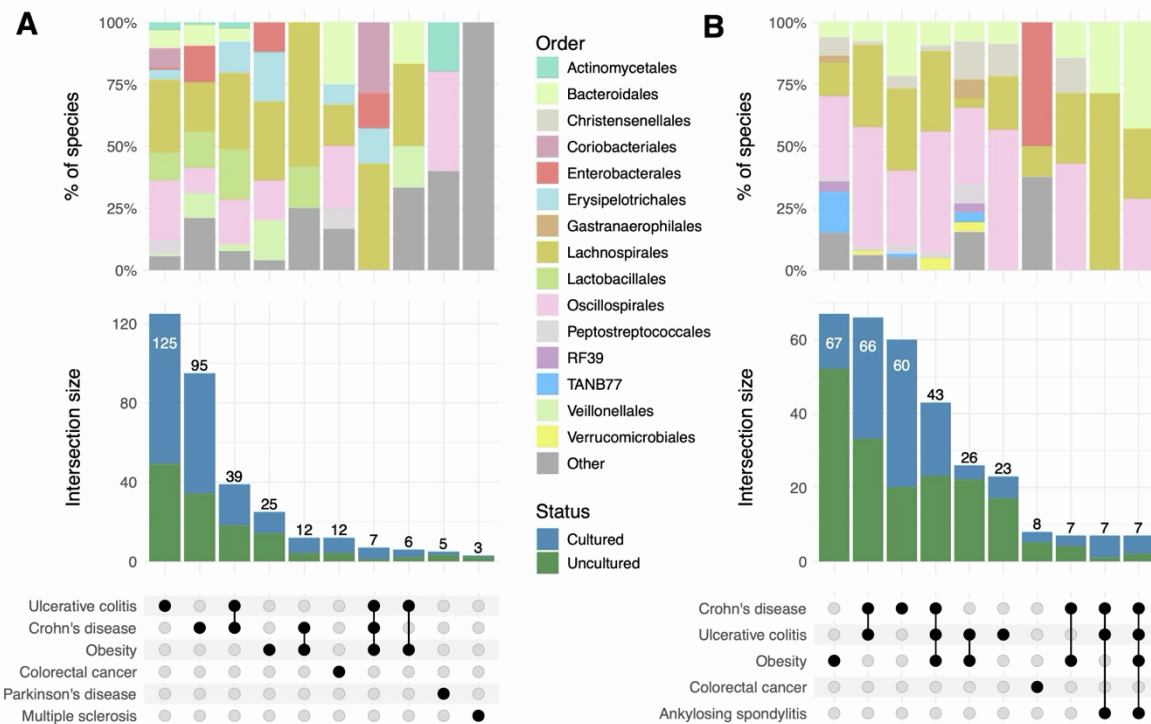


Figure S6. Overlap of health- and disease-associated species across multiple conditions.

Related to Figure 2.

(A) Number of prokaryotic species classified as disease associated (overrepresented in disease) and the level of overlap across multiple conditions. Vertical bars represent the number of significant species found in the diseases highlighted with black dots in the lower panel (only the top 10 intersections are shown). Stacked bar plots at the top display the taxonomic affiliation (order rank) of the species within each intersection.

(B) Same representation as panel (A) but only considering health-associated species (overrepresented in healthy controls).

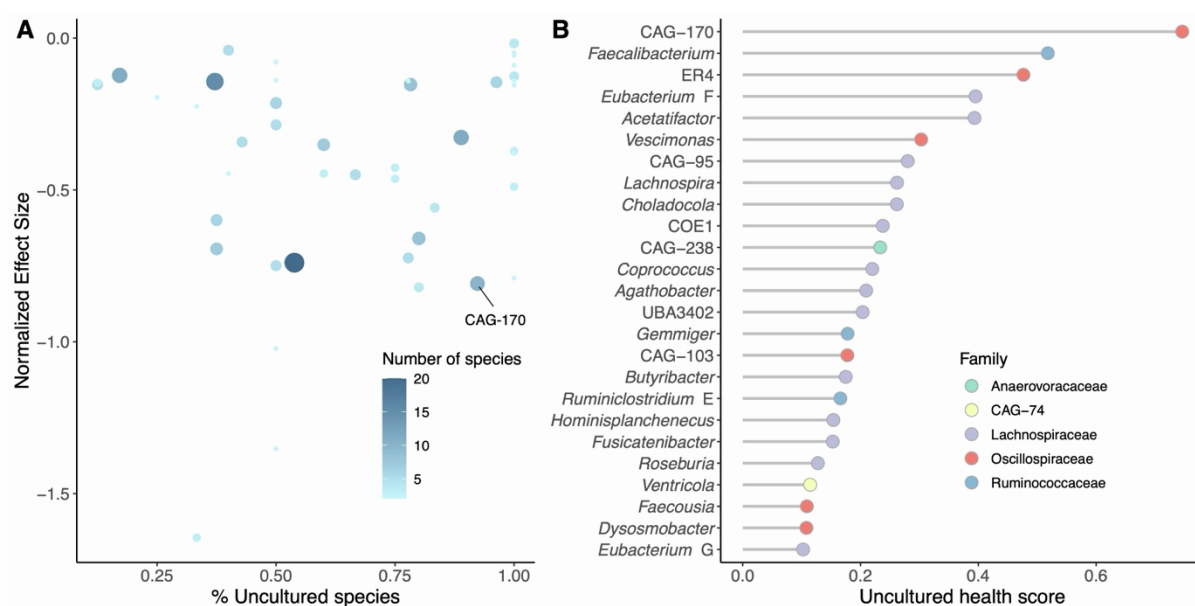


Figure S7. Validation of uncultured health scores using the Stouffer's method. Related to Figure 3.

(A) Proportion of uncultured species relative to the estimated effect size for each bacterial genus in the UHGG catalog. A negative effect size indicates an association with health. The size and colour of each data point (genus) is scaled according to the number of health-associated species identified (only genera with an overall association with health are shown). Effect sizes and P values for defining statistical significance were combined across diseases using the Stouffer's method.

(B) Bacterial genera ranked by the absolute value of their uncultured health score, calculated as the product of the number of significant species ($\log_{10}$ -transformed), normalized effect size, and proportion of uncultured species.

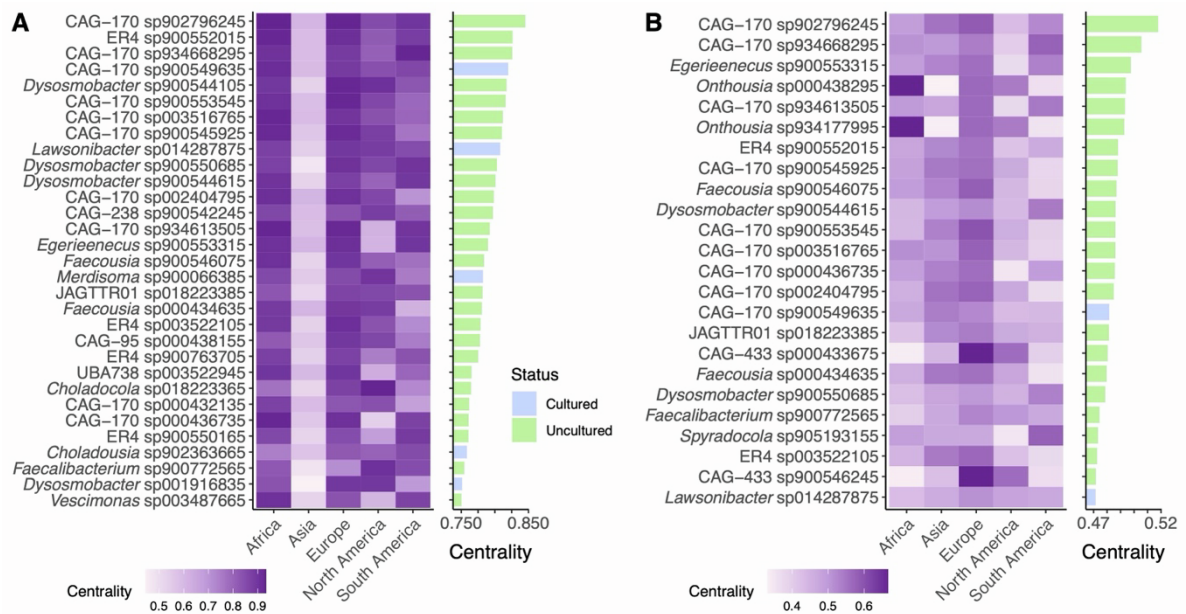


Figure S8. Species classified as network hubs following additional correlation filters.

Related to Figure 4.

(A) Centrality scores of the top 1% network hubs after filtering network edges with a minimum $|\text{correlation}|$ of 0.1. The heatmap shows continent-specific values, and the bar plot shows the mean centrality score across continents.

(B) Same representation as panel (A) but with networks filtered with a minimum $|\text{correlation}|$ of 0.2. The genus CAG-170 is the most represented taxon within both sets.

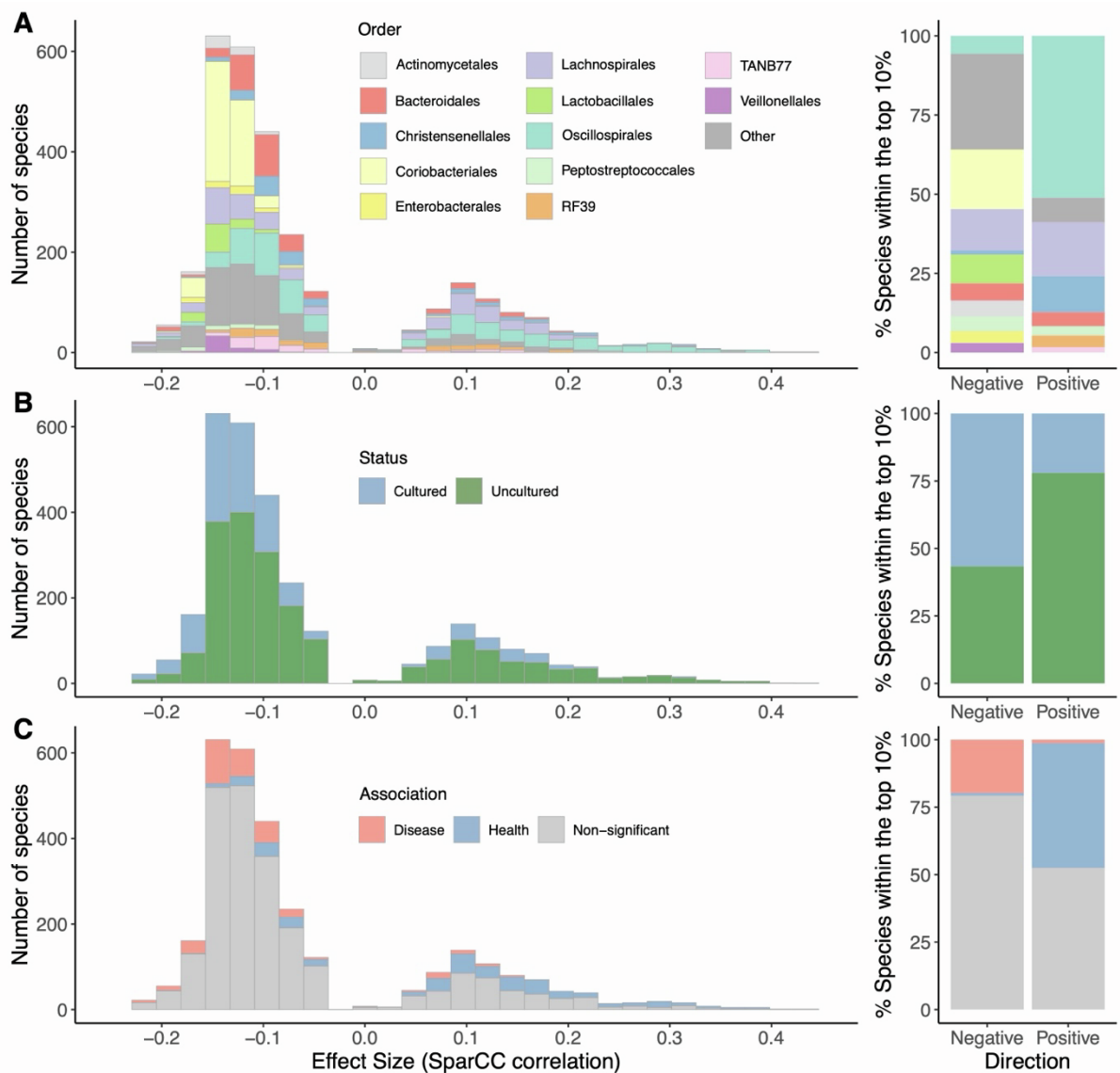


Figure S9. Gut microbiome species positively or negatively associated with CAG-170. Related to Figure 4.

(A) Distribution of the number of species positively (correlation > 0) or negatively (correlation < 0) associated with CAG-170 in the SparCC networks. Correlation values represent the median across continents and across all CAG-170 species. Only correlations found to be significant ($FDR < 0.05$) are represented. Bars are coloured based on taxonomic affiliation (order rank). The right panel denotes the taxonomic distribution of the top 10% negative and top 10% positive species.

(B) Same representation as panel (A) but with species coloured based on their cultured status (“cultured” or “uncultured”).

(C) Same representation as panel (A) but with species coloured based on the health association identified in this study (health associated, disease associated or non-significant).

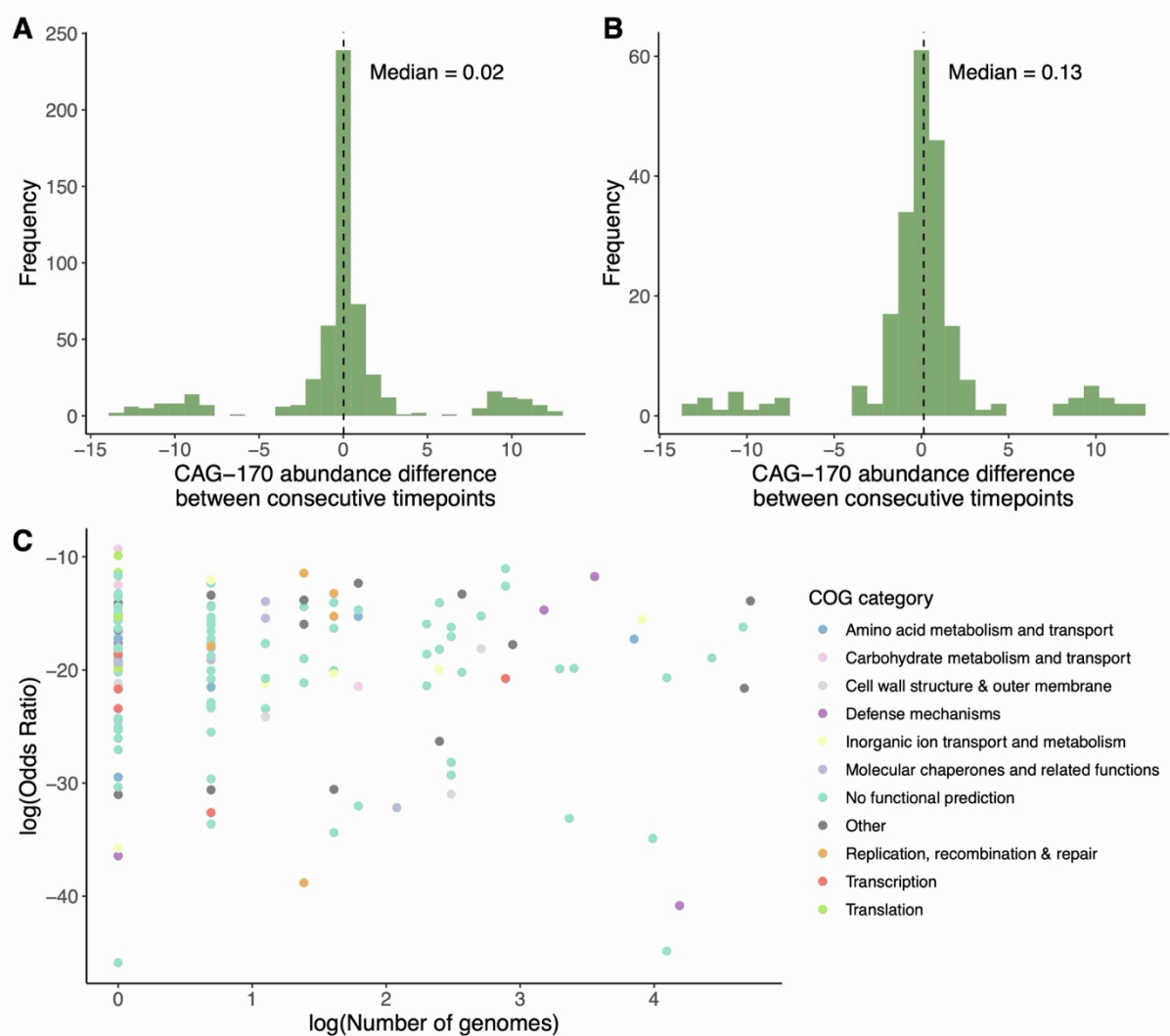


Figure S10. CAG-170 stability and genetic diversity in a longitudinal cohort. Related to Figure 5.

(A) Distribution of CAG-170 abundance differences between consecutive timepoints per subject in the HMP2-IBD study. Abundance was measured using centred log-ratio transformed counts. Vertical line denotes the median value. Only subjects in which CAG-170 was detected in at least one timepoint were considered.

(B) Same representation as (A) but only considering subjects in which CAG-170 was detected in more than 10 timepoints.

(C) Representation of the 160 CAG-170 genes negatively associated with dysbiosis, relating the number of high-quality genomes containing each gene with the corresponding effect size (log-transformed odds ratio). Genes are coloured by their Clusters of Orthologous Groups (COG) functional category. Effect sizes and *P* values were calculated using a mixed-effects linear model that accounted for repeated measures within subjects.

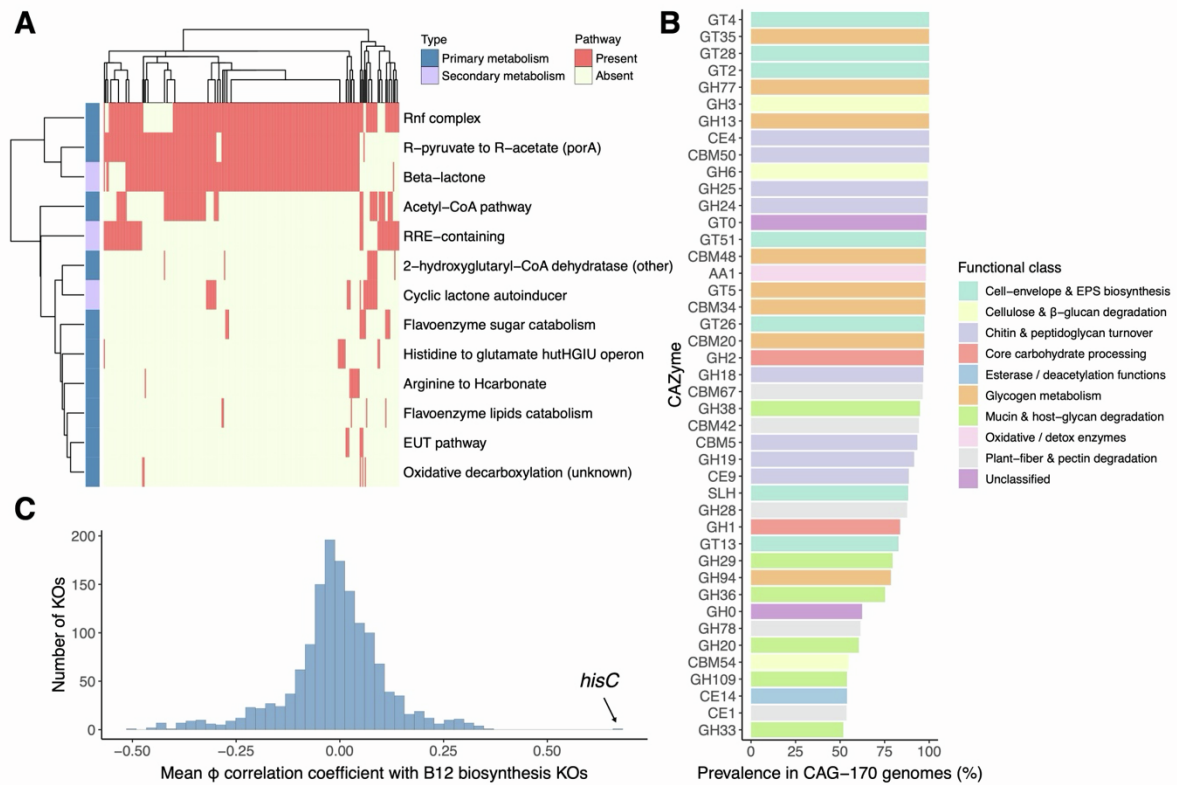


Figure S11. Prediction of CAG-170 core functional repertoire and B12 association.

Related to Figure 6.

(A) Prevalence patterns of metabolic pathways (rows) predicted among all high-quality CAG-170 genomes ($n = 332$ genomes, depicted as columns). Primary metabolic pathways were predicted with gutSMASH and secondary metabolites with antiSMASH.

(B) Most prevalent CAZymes detected among CAG-170 genomes. Each CAZyme is coloured by a broad functional class inferred from manual curation of existing literature.

(C) Histogram denoting the number of KEGG Orthologs (KOs) and their correlation (mean ϕ correlation coefficient) with B12 biosynthesis KOs. The KO with the strongest association (highest coefficient) encoding a histidinol-phosphate aminotransferase (*hisC* gene) is indicated with an arrow.

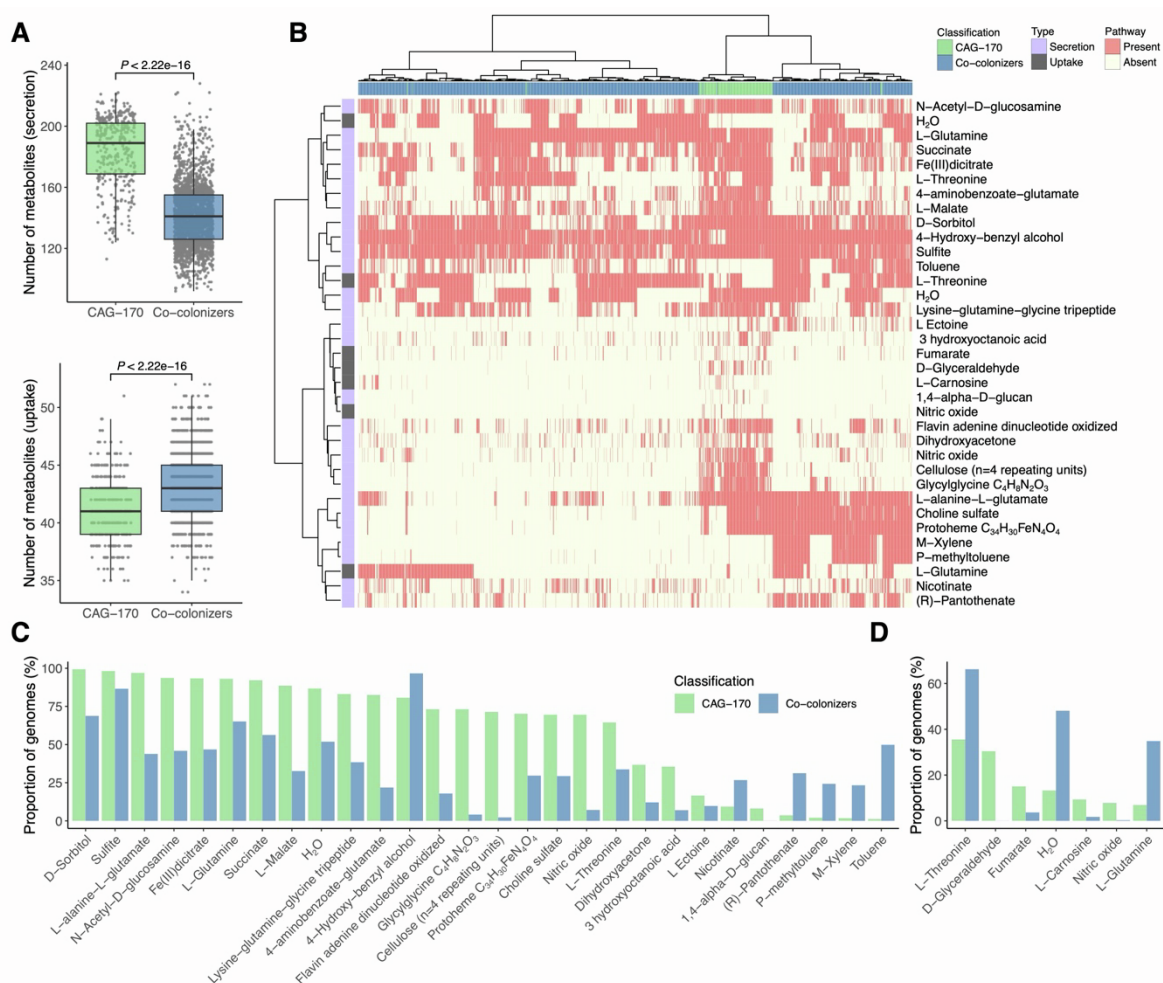


Figure S12. Metabolic relationship between CAG-170 and its co-colonizers. Related to STAR Methods.

(A) Comparison of the number of metabolites predicted from secretion (top) and uptake (bottom) fluxes between CAG-170 ($n = 332$ genomes) and its top 1% positive network members ('co-colonizers', $n = 2,006$ genomes). P values were derived from a two-sided Wilcoxon rank-sum test. Box lengths represent the IQR of the data, the central line represents the median value, and the whiskers depict the lowest and highest values within 1.5 times the IQR of the first and third quartiles, respectively.

(B) Metabolites significantly associated with either CAG-170 or its co-colonizers ($\text{FDR} < 0.05$) based on a mixed-effects logistic regression analysis. Metabolites (rows) and genomes (columns) are clustered using the Ward's method based on their presence/absence patterns.

(C) Prevalence of the metabolites predicted from secretion fluxes found to be significantly different between CAG-170 and its co-colonizers. Values represent the proportion of genomes within each group predicted to secrete each metabolite under a rich gut media (M3).

(D) Same representation as (C) but for metabolites predicted from uptake fluxes.